

MATRIX STIFFNESS ANALYSIS: FRAMES AND GRILLAGES

In this chapter we consider structures which are made up of one-dimensional beam or bar elements connected together at their ends. The connection may be either pinned or rigid. Such structures are referred to as frames and the connection points are referred to as *nodes*. Frames are probably the most common class of structure. In a ship, in spite of the plating, the three-dimensional assemblage of deck beams, side frames, and longitudinal girders constitutes a framework, especially in regard to the transverse loads because these act normal to the plating and are carried mainly by the framing system.

5.1 BASIC CONCEPTS

Frame Analysis: Nodal Displacements

Frame analysis is a well established technique in the field of structural analysis. In this technique everything is expressed in terms of what happens at the nodes. External forces are applied at the nodes (and only there), and the displacements of the structure are expressed entirely in terms of nodal displacements. The starting point is to determine the response characteristics of each individual member, that is, the relationship between nodal forces and nodal displacements. This relationship is taken to be linear, which means that each nodal displacement u_j is linearly proportional to each nodal force f_i : $f_i = k_{ij} u_j$. The coefficient k_{ij} is referred to as a *stiffness coefficient*. The complete relationship between all nodal forces and all nodal displacements in a member is a system of

linear equations, and such systems are best expressed in matrix notation, $\mathbf{f} = \mathbf{k}\mathbf{u}$, in which $\mathbf{f}$ and $\mathbf{u}$ are vectors containing the nodal forces and displacements, and $\mathbf{k}$ is a square matrix containing all of the stiffness coefficients; this is the *stiffness matrix* of the member.

Fundamental Laws

In structural analysis there are three fundamental "laws" or relationships which must be satisfied:

1. Equilibrium of forces (within each member and between members).
2. Compatibility of displacements (within each member and between members).
3. Law of material behavior (stress-strain law) of each member.

As noted, the first two of these must be satisfied at two levels: within each member and also for the structure as a whole (i.e., between members). For one-dimensional members such as bars and beams, which we deal with in this chapter, all three laws are exactly satisfied within each member because they are implicit in the member's nodal force-nodal displacement relationship, that is, the member stiffness. Therefore, this chapter deals mainly with the first two of the laws applied to the structure as a whole. In other words, there must be equilibrium of forces between the external loads and the various member forces, and there must be compatibility in the deformations of the members, such that they continue to fit together.

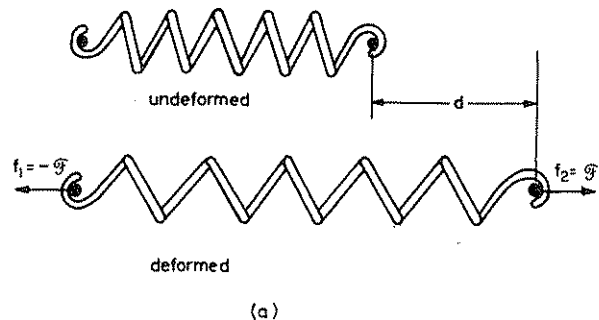
For a statically determinate structure the equilibrium requirement is sufficient because the member forces can be calculated directly, whereupon the member displacements and internal forces can be calculated directly. But a frame structure is statically indeterminate, and so in frame analysis the equilibrium and compatibility requirements are imposed at each node of the structure. These requirements, together with the force-displacement relationships of each member (as embodied in its stiffness matrix $\mathbf{k}$) produce a system of equations for the nodal displacements. After solving for these displacements, some previously established relationships between them and the member's internal forces and deformations (that is, stresses and strains) are used to solve for the latter.

In a typical three-dimensional frame structure there may be a large number of nodal displacements, several thousand perhaps, but the solution of a large system of linear equations is a routine task for a computer. Because there are a large number of equations, any discussion of the underlying theory and of the setting up of the equations is greatly facilitated by using matrix notation; hence this field is often referred to as matrix frame analysis or matrix stiffness analysis.

For one-dimensional members, imposing the requirements of equilibrium and compatibility at the nodes is sufficient to ensure that they are satisfied everywhere, both within members and between members, because the members are connected only at the nodes. For members that are two- or three-dimensional in extent it is necessary to define a relationship between nodal displacements and internal deformation in such a way that these two laws are satisfied to sufficient accuracy both within the member and between members. This is the key step in finite element analysis, and it is the only major difference between the basic finite element method and frame analysis. Hence much of the material of this chapter serves as a foundation for the presentation of the finite element method in Chapter 7.

Stiffness Matrix of a Structure

The device most directly associated with stiffness is a simple elastic spring, such as that shown in Fig. 5.1. It can also be regarded as an example of a structural member because it has the same basic structural characteristics as a pin-ended bar: it undergoes axial displacements, transmits axial forces, and exhibits a linear internal (or material) behavior; that is, there is a linear relationship between internal force $\mathcal{F}$ (axial



(a) Nodal forces f_1 (negative; $f_1 = -\mathcal{F}$) and f_2 (positive; $f_2 = \mathcal{F}$)
 (b) Internal force $\mathcal{F}$. Magnitude of $\mathcal{F}$ = magnitude of a pair of equal and opposite nodal forces required to cause a specified deformation, d . Sign of $\mathcal{F}$: positive for elongation.

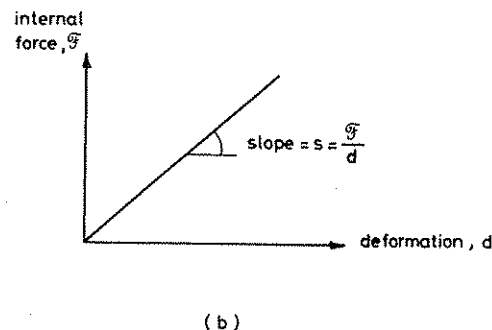


Figure 5.1 Internal force–deformation relationship for an elastic spring element.

force) and internal deformation d (elongation or shortening) which is of the form

$$\mathcal{F} = sd$$

The quantity s is termed the *stiffness* of the spring and corresponds to the slope of the internal force–deformation diagram (Fig. 5.1).*

Besides being a structural member the same spring could be considered as a structure—a one-member structure which is supported at one node and loaded at the other by an applied load F (see Fig. 5.2). The load

*Note that there is a subtle but important distinction between the internal force in a member and the applied external forces f_1 and f_2 . Although their definitions are such that they always have the same magnitude, they are conceptually quite different. The internal force $\mathcal{F}$ is a state or situation inside of the member, and it is defined in terms of the internal deformation. In the present case it is the magnitude which a pair of equal and opposite forces must have in order to cause a specified elongation. It is a scalar quantity (no direction), it is diffused throughout the member, and its sign depends on the type of deformation: positive for elongation. In contrast the external forces are directional and can be regarded as vectors (although here they have only one component each), they act at specific locations, and their sign depends on their direction.

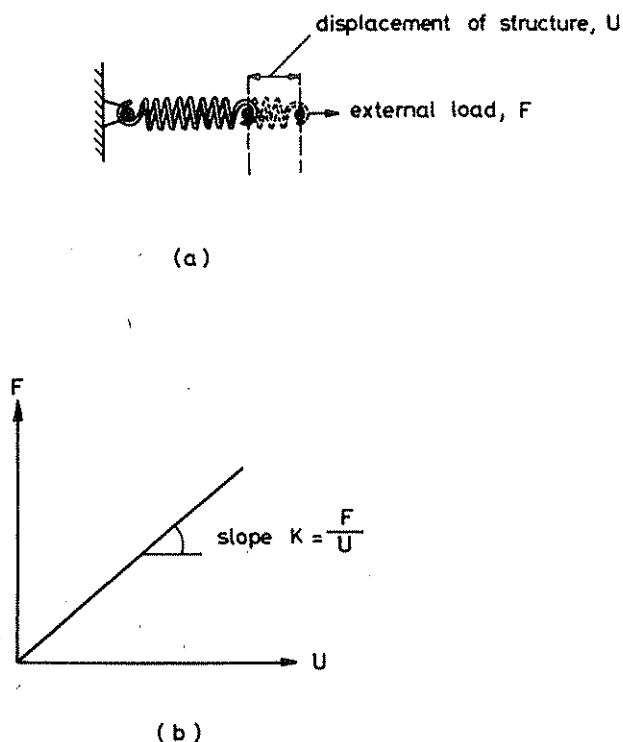


Figure 5.2 Load-displacement relationship for an elastic spring structure.

causes a structure displacement U which is linearly proportional to F ,

$$F = KU \quad (5.1.1)$$

and the constant of proportionality K is termed the stiffness of the structure. In this case, since the structure is identical to the member, the two stiffnesses are the same: $K = s$. Knowing the value of this stiffness and of the applied load, (5.1.1) may be inverted to give the displacement

$$U = \frac{1}{K} F \quad (5.1.2)$$

For this structure the response consists of just one displacement. For more realistic structures such as the pin-jointed frame shown in Fig. 5.3, it is necessary to determine the displacements of nodes C, D, E, and F in order to be able to evaluate the member deformations (axial strains) and hence the internal forces (stresses) in the members.

Thus, the most basic task of structural analysis is to determine the relationship between the nodal loads and the nodal displacements of the structure. This relationship will be analogous to (5.1.1) for the simple spring structure, but there will be many inter-

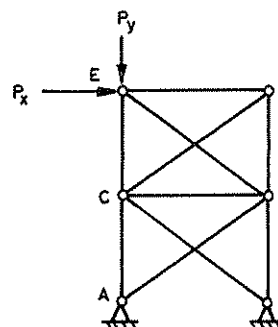


Figure 5.3 Statically indeterminate pin-jointed frame.

connecting members and therefore many interrelated nodal displacements. Likewise, there can be many external loads because these may occur at any node. Therefore a general linear relationship between all of the external loads and all of the nodal displacements will consist of a system of simultaneous equations, and such systems are best expressed in matrix notation. The load-displacement relation is therefore of the form

$$\mathbf{F} = \mathbf{K}\mathbf{U} \quad (5.1.3)$$

in which $\mathbf{F}$ and $\mathbf{U}$ are vectors of nodal loads and nodal displacements respectively, and the matrix $\mathbf{K}$ is termed the "stiffness matrix" of the structure, since it consists of the coefficients relating $\mathbf{F}$ to $\mathbf{U}$. For the very simple structure of Fig. 5.2, which has only one possible displacement, this matrix is of order 1×1 and the vectors $\mathbf{F}$ and $\mathbf{U}$ contain only one term each.

It can now be seen that the basic task of determining the load-displacement relationship of a structure consists essentially in obtaining the stiffness matrix of the structure because once $\mathbf{K}$ has been determined, the solution for $\mathbf{U}$ follows immediately. As will now be shown, the load-displacement relationship for each member or *element* (we will use the latter term from now on) can also be expressed in terms of an element stiffness matrix $\mathbf{k}^e$. Next, it will be shown that the structure stiffness matrix $\mathbf{K}$ is in fact obtained ("assembled") by a systematic superposition of the element stiffness matrices, taking account of the particular way in which the elements are arranged and connected in the structure.

Stiffness Matrix of a Spring (or Bar) Element

For simplicity we again consider the simple spring of Fig. 5.1 and we now regard it as a basic element, forming part of a larger structure. In this case each of its two nodes would be connected to other elements in

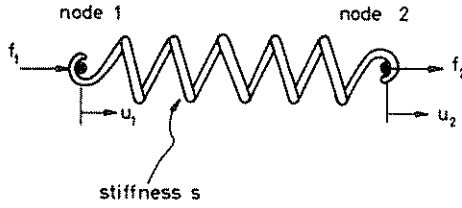


Figure 5.4 Element nodal forces and displacements for the spring element.

the structure and so in general each node would be subjected to a nodal force and would undergo a nodal displacement, as shown in Fig. 5.4. For a pin-jointed bar element the stiffness s is AE/L . Positive forces and displacements are as shown in the figure. The vector of nodal forces is

$$\mathbf{f} = \begin{Bmatrix} f_1 \\ f_2 \end{Bmatrix}$$

and the vector of nodal displacements is

$$\mathbf{u} = \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix}$$

Therefore the stiffness matrix for the spring element is of order 2×2 and the relationship between nodal forces and nodal displacements is of the form

$$\begin{Bmatrix} f_1 \\ f_2 \end{Bmatrix} = \begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} \quad \text{or} \quad \mathbf{f} = \mathbf{k}^e \mathbf{u} \quad (5.1.4)$$

in which the superscript e indicates that this stiffness matrix is for a single element. We now seek to determine the individual terms of $\mathbf{k}^e$. These can be obtained by considering each nodal displacement in isolation (i.e., keeping the other nodal displacement zero) and using the law of material behavior to evaluate the two nodal forces. For example, let us consider nodal displacement u_1 and keep $u_2 = 0$, as shown in Fig. 5.5a. From equilibrium the two nodal forces must be equal and opposite, and from the definition of $\mathcal{F}$ their magnitude must be the same as that of $\mathcal{F}$. Since $u_2 = 0$ the deformation is $d = u_2 - u_1 = -u_1$, and hence the material law of the spring requires that $\mathcal{F} = s(-u_1)$. Therefore $f_1 = -\mathcal{F} = su_1$ and $f_2 = -f_1 = -su_1$. If these values are substituted into (5.1.4) the result is

$$\begin{Bmatrix} su_1 \\ -su_1 \end{Bmatrix} = \begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 = 0 \end{Bmatrix}$$

from which

$$su_1 = k_{11}u_1 \quad \text{or} \quad k_{11} = s$$

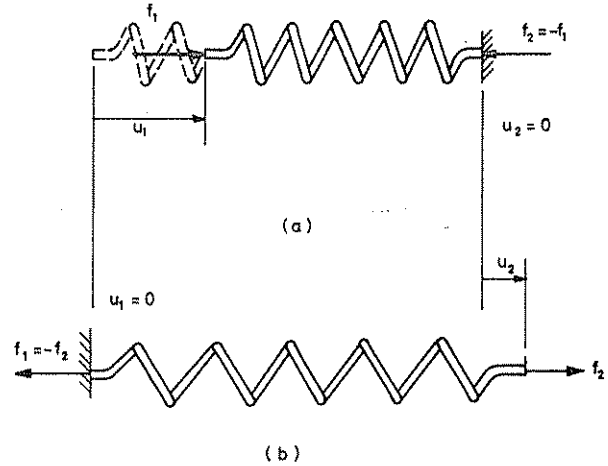


Figure 5.5 Calculation of $\mathbf{k}^e$ by imposing individual nodal displacements.

and

$$-su_1 = k_{21}u_1 \quad \text{or} \quad k_{21} = -s$$

We have thus obtained the terms in the first column of $\mathbf{k}^e$. The terms in the second column are obtained in a similar manner by setting $u_1 = 0$ (Fig. 5.5b) and by again observing that from equilibrium and the definition of the spring constant s the nodal forces are $f_1 = -su_2$ and $f_2 = su_2$. Substituting these values into (5.1.4) and again noting that $u_1 = 0$ gives

$$-su_2 = k_{12}u_2 \quad \text{or} \quad k_{12} = -s$$

together with

$$su_2 = k_{22}u_2 \quad \text{or} \quad k_{22} = s$$

Therefore the complete element stiffness matrix for the spring element is

$$\mathbf{k}^e = \begin{bmatrix} s & -s \\ -s & s \end{bmatrix} \quad (5.1.5a)$$

Hence the relationship between element nodal forces and element nodal displacements is

$$\begin{Bmatrix} f_1 \\ f_2 \end{Bmatrix} = \begin{bmatrix} s & -s \\ -s & s \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} \quad \text{or} \quad \mathbf{f} = \mathbf{k}^e \mathbf{u} \quad (5.1.5b)$$

It may be seen that the stiffness matrix is symmetric. This is true of all stiffness matrices, whether for a single element or for an entire structure. The symmetry comes from (or is an alternative way of stating) Maxwell's reciprocal theorem. Further examination of

k^e would also reveal that it is singular; the reason for this will be explained subsequently.

It may also be noted that in obtaining k^e we made use of the laws of material behavior and of equilibrium within the element. For such a simple element the law of compatibility hardly arises but it *is* satisfied; the spring's internal deformation (stretching or shortening) is linear and continuous.

Assembling the Structure Stiffness Matrix

The next point to consider is how the stiffness matrix for a structure can be obtained from the element stiffness matrices of its constituent elements. To achieve this we make use of the element force-displacement relationship, (5.1.5), of which the element stiffness matrix is the crucial part, together with the two laws that must be met at the structure level—those of equilibrium and of compatibility. Both laws are to be applied at the nodes of the structure (because everything is being described in terms of nodal values) and so we need a structure node numbering system, quite apart from the element node numbering system of Fig. 5.4. We will also need a sign convention for the applied loads F and for the structure nodal displacements U .

Consider, for example, the two-element structure of Fig. 5.6. The structure nodes are labeled A, B, and C (letters are chosen instead of numbers to emphasize the distinction between the element system and the structure system) and both the loads F and the displacements U are positive to the right. Since there are three nodes there may be up to three applied (external)

loads, as shown in the figure, but there could just as well be only one or two.*

Also, each load will act in a particular direction. In the example loads F_B and F_C act towards the right and load F_A to the left. Hence the numerical value of the latter would be negative.

The law of equilibrium requires that at each node there must be equilibrium between the external applied load at that node (if any) and the sum of the element forces at that node. Therefore (see Fig. 5.6)

$$F_A = f_{a1}$$

$$F_B = f_{a2} + f_{b1}$$

$$F_C = f_{b2}$$

Next, we use the element force-displacement relationship, that is, the element stiffness expressions of (5.1.5), to express the element nodal forces f in terms of the element nodal displacements u . The foregoing equations then become

$$F_A = s_a u_{a1} - s_a u_{a2}$$

$$F_B = -s_a u_{a1} + s_a u_{a2} + s_b u_{b1} - s_b u_{b2}$$

$$F_C = -s_b u_{b1} + s_b u_{b2}$$

The next step is to impose interelement compatibility of the *structure* nodal displacements. This consists essentially in relating the element displacements u to the structure displacements U . In this example the relationships are

$$u_{a1} = U_A$$

$$u_{a2} = u_{b1} = U_B$$

$$u_{b2} = U_C$$

Substituting these gives

$$F_A = s_a U_A - s_a U_B$$

$$F_B = -s_a U_A + (s_a + s_b) U_B - s_b U_C$$

$$F_C = -s_b U_B + s_b U_C$$

which are the desired equations relating the applied loads to the structure displacements. Therefore the coefficients of these equations constitute the individual terms of the structure stiffness matrix K , introduced in

*For the structural analysis the applied loads are known quantities, although it will often be the task of the structural analyst to calculate them before commencing the analysis.

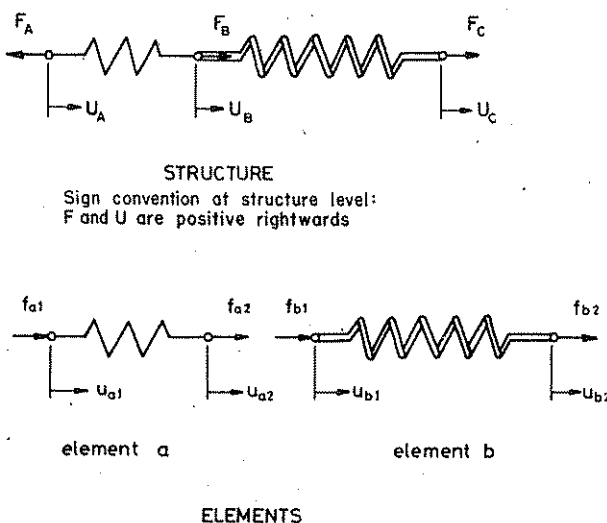


Figure 5.6 Two-element structure.

(5.1.3). In matrix form the above equations are

$$\begin{Bmatrix} F_A \\ F_B \\ F_C \end{Bmatrix} = \begin{bmatrix} s_a & -s_a & 0 \\ -s_a & s_a + s_b & -s_b \\ 0 & -s_b & s_b \end{bmatrix} \begin{Bmatrix} U_A \\ U_B \\ U_C \end{Bmatrix}$$

or $\mathbf{F} = \mathbf{K}\mathbf{U}$ (5.1.6)

It may be seen that $\mathbf{K}$ is an assemblage which includes each and every one of the individual terms of $\mathbf{k}^a$ and $\mathbf{k}^b$, assembled together in a systematic manner. Hence the laws of equilibrium and compatibility are equivalent to a rule or procedure by which the terms of the $\mathbf{k}^e$ matrices are assembled to form $\mathbf{K}$. If we can specify this procedure in a suitably general and practical form we can then obtain $\mathbf{K}$ by a direct assembly process, without having to explicitly impose these laws. To achieve this we first show that the assembly process is essentially a superposition of the element stiffness matrices, and then we present a procedure for assembling $\mathbf{K}$ which is computationally practical and efficient.

To demonstrate the superposition of the element stiffness matrices we first write down the full set of equilibrium equations for each element.

$$\begin{array}{cc} \text{Element } a & \text{Element } b \\ \begin{Bmatrix} f_A \\ f_B \\ f_C \end{Bmatrix} = \begin{bmatrix} s_a & -s_a \\ -s_a & s_a \end{bmatrix} \begin{Bmatrix} U_A \\ U_B \end{Bmatrix} & \begin{Bmatrix} f_B \\ f_C \end{Bmatrix} = \begin{bmatrix} s_b & -s_b \\ -s_b & s_b \end{bmatrix} \begin{Bmatrix} U_B \\ U_C \end{Bmatrix} \end{array}$$

Although the two $\mathbf{k}^e$ matrices are of the same order they may not be added directly since they relate to different sets of displacements. However, by inserting rows and columns of zeros, both may be expanded in such a way that they each relate to the three displacements U_A , U_B , and U_C .

$$\begin{Bmatrix} f_A \\ f_B \\ f_C \end{Bmatrix} = \begin{bmatrix} s_a & -s_a & 0 \\ -s_a & s_a & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} U_A \\ U_B \\ U_C \end{Bmatrix}$$

$$\begin{Bmatrix} f_A \\ f_B \\ f_C \end{Bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & s_b & -s_b \\ 0 & -s_b & s_b \end{bmatrix} \begin{Bmatrix} U_A \\ U_B \\ U_C \end{Bmatrix}$$

We now impose equilibrium of forces which says that the sum of the element forces at each node is equal to the external load at that node (or equals zero if there is no external load). Performing the summation gives

$$\begin{Bmatrix} F_A \\ F_B \\ F_C \end{Bmatrix} = \begin{bmatrix} s_a & -s_a & 0 \\ -s_a & s_a + s_b & -s_b \\ 0 & -s_b & s_b \end{bmatrix} \begin{Bmatrix} U_A \\ U_B \\ U_C \end{Bmatrix}$$

From this we can see that the way in which the individual terms of $\mathbf{k}^a$ and $\mathbf{k}^b$ are combined together (assembled) to form $\mathbf{K}$ depends entirely on how the physical elements are arranged (or assembled) in the actual structure. In explaining the assembly procedure in more detail we shall identify each of the individual terms in a stiffness matrix by its row and column, and we note here that in $\mathbf{K}$ these rows and columns correspond to structure nodes, whereas in $\mathbf{k}^a$ and $\mathbf{k}^b$ they correspond to element nodes. Also, the arrangement of the elements in the structure may be described by specifying the structure node where each element node is located. That is, for each node of each element the corresponding structure node is specified, and this summarizes the basic layout of the structure. For this example the structure node corresponding to each element node is shown in Fig. 5.7 immediately below the element stiffness matrices.

The foregoing summation of matrices shows that each term in a particular row and column position of $\mathbf{K}$ (corresponding to structure nodes I and J , say) is the sum of all of the $\mathbf{k}^e$ terms whose row and column numbers (i.e., whose element node numbers) correspond to (are located at) structure nodes I and J . Thus a particular $\mathbf{k}^e$ term, say k_{lm} , is added into the K_{IJ} position of $\mathbf{K}$, where I and J are the structure nodes at which the element nodes l and m are located in the structure.

With this rule it is no longer necessary to expand each of the element matrices up to the full structure size and then perform matrix addition. Instead, we may think of each of the element matrices as being "broken up" and its individual terms being inserted directly into their proper position in $\mathbf{K}$, according to an "assembly plan" which is simply a list of the structure nodes corresponding to the element nodes of that element. With this procedure, the only information which is required is the list of structure node numbers for each element. This list is usually referred to as the *location vector*. There is one of these for each element and it is usually a very short vector since its length equals the number of nodes in the element. The location vectors for the above example are shown in Fig. 5.7.

Because each element stiffness matrix is symmetric the resulting structure stiffness matrix is likewise always symmetric. Therefore only half of it needs to be assembled.

In the simple example presented here the elements are all in-line and there is only one component of displacement at each node. With more practical elements there are several components of displacement at each node and so in later sections of this chapter, and

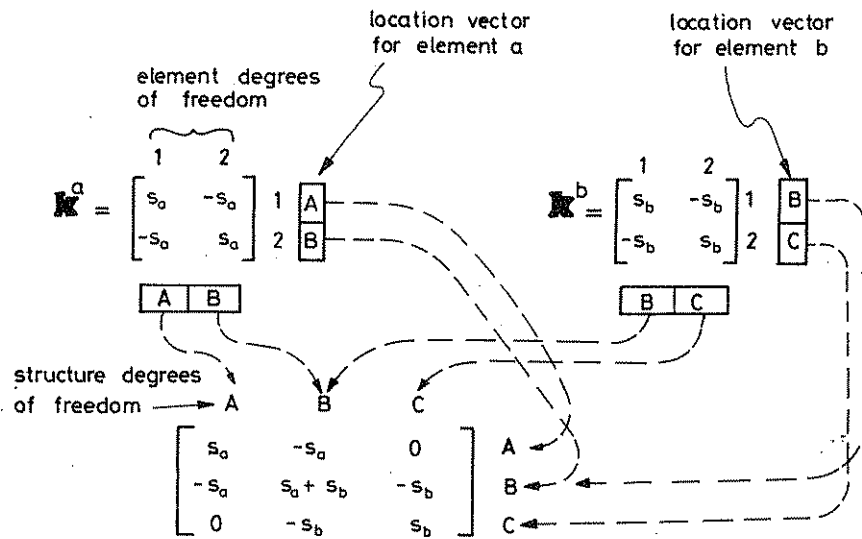


Figure 5.7 Location vectors and assembly of the structure stiffness matrix.

also in Chapter 7, the foregoing definitions will need to be generalized. Nevertheless these simple in-line spring elements illustrate all of the basic principles of matrix stiffness analysis and provide a useful introduction.

Solution Procedure

After the structure stiffness matrix has been fully assembled an apparent paradox arises—in every case the matrix is singular! This would seem to indicate that the equations for the structure's nodal displacements cannot be solved; that the displacements are indeterminate. This apparent paradox has a simple explanation—the structure has not yet been tied down or given any points of support, and in this situation the applied loads would cause the entire structure to move as a rigid body, thus indeed giving indeterminate displacements. Obviously, then, it is necessary to provide sufficient restraints on the structure to prevent this rigid body motion. These restraints are applied by specifying zero values for some of the structure's displacements. In practice structures often have more restraints than the minimum number. If so, they must all be specified because otherwise the structure would not be accurately represented. The full set of restraints is referred to as the support conditions. In the simple two-element structure of Fig. 5.6 there is only one possible type of rigid body motion—horizontal movement—and hence only one restraint is required for this structure. For example, let us say that the structure is

supported at node A, so that $U_A = 0$. Then (5.1.6) can be rewritten in partitioned form as follows:

$$\begin{Bmatrix} F_A \\ F_B \\ F_C \end{Bmatrix} = \begin{bmatrix} s_a & -s_a & 0 \\ -s_a & s_a + s_b & -s_b \\ 0 & -s_b & s_b \end{bmatrix} \begin{Bmatrix} U_A = 0 \\ U_B \\ U_C \end{Bmatrix} \quad (5.1.7)$$

The system of equations contains two unknown displacements U_B and U_C , and an unknown reaction F_A . F_B and F_C are known applied loads. Using standard matrix manipulation we have

$$\begin{aligned} \{F_A\} &= s_a \{U_A\} + [-s_a \ 0] \begin{Bmatrix} U_B \\ U_C \end{Bmatrix} \\ \begin{Bmatrix} F_B \\ F_C \end{Bmatrix} &= \begin{Bmatrix} -s_a \\ 0 \end{Bmatrix} \{U_A\} + \begin{bmatrix} s_a + s_b & -s_b \\ -s_b & s_b \end{bmatrix} \begin{Bmatrix} U_B \\ U_C \end{Bmatrix} \end{aligned}$$

and since U_A is zero we have

$$\{F_A\} = [-s_a \ 0] \begin{Bmatrix} U_B \\ U_C \end{Bmatrix} \quad (5.1.8)$$

together with

$$\begin{Bmatrix} F_B \\ F_C \end{Bmatrix} = \begin{bmatrix} s_a + s_b & -s_b \\ -s_b & s_b \end{bmatrix} \begin{Bmatrix} U_B \\ U_C \end{Bmatrix} \quad (5.1.9)$$

Equation 5.1.9 consists of two equations in the two unknowns U_B and U_C and may be solved for these values which, when substituted in (5.1.8) then give the value of the unknown reaction F_A .

It should be noted that (5.1.9) may be obtained directly from (5.1.7) simply by deleting the rows and columns of $\mathbf{K}$ corresponding to zero displacements.

Once the displacements have been obtained, the internal forces in the elements may be determined from the law of material (or internal) behavior of the element, that is, the relationship between internal force and internal deformation. For a spring this relationship is simply $\mathcal{F} = sd$ where $\mathcal{F}$ is the internal force in the spring and d is the change in length of the spring. Hence for the two elements we have

$$\begin{aligned}\mathcal{F}_a &= s_a(u_{a2} - u_{a1}) \\ \mathcal{F}_b &= s_b(u_{b2} - u_{b1})\end{aligned}\quad (5.1.10)$$

Since the elements are all in-line the required element displacements may be obtained by simply substituting the appropriate structure displacements.

$$\begin{aligned}\mathcal{F}_a &= s_a(U_B - U_A) \\ \mathcal{F}_b &= s_b(U_C - U_B)\end{aligned}$$

This completes the solution process. It will be helpful at this point to summarize the principal steps in matrix stiffness analysis because these remain basically the same regardless of the type of structure and the types of element which are used. The seven principal steps are given in Table 5.1. The transformation referred to in Step 2 was not necessary in the foregoing example. It is introduced and explained in the next section. Step 7 will be explained in Section 5.3.

TABLE 5.1 PRINCIPAL STEPS IN MATRIX STIFFNESS ANALYSIS

- 1 Define the structural model (structure coordinate system, nodes, node numbers, element types, and loads).
- 2 Determine each element stiffness matrix $\mathbf{k}^e$ and transform it to structure coordinates (whereupon it is denoted $\mathbf{K}^e$).
- 3 Assemble the structure stiffness matrix $\mathbf{K}$ from the individual element stiffness matrices $\mathbf{K}^e$ thus obtaining the system of equilibrium equations.
- 4 Apply the support conditions to the system of equations.
- 5 Solve for the displacements and then, if desired, the reaction forces at the supports.
- 6 For each element, obtain the nodal displacements and then use the internal force (or stress) matrix of that element to obtain the internal forces (or stresses).
- 7 If equivalent nodal loads were used to represent distributed loads, superimpose the fixed-end forces from the solution of the "internal problem."

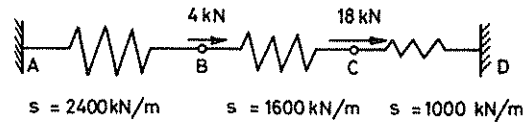


Figure 5.8 Numerical example, three-element structure.

Numerical Example

A simple numerical example is given to illustrate the procedure. The structure shown in Fig. 5.8 consists of three springs and is supported at nodes A and D. If axial loads of 4 kN and 18 kN are applied at nodes B and C respectively as shown, determine the displacements at nodes B and C and the reaction forces at A and D.

The element stiffness matrices and location vectors are:

$$\text{Element a: } \mathbf{K}^a = \begin{array}{c|c} & \begin{array}{cc} \text{A} & \text{B} \end{array} \\ \hline \begin{array}{c} \text{A} \\ \text{B} \end{array} & \begin{bmatrix} 2400 & -2400 \\ -2400 & 2400 \end{bmatrix} \end{array}$$

$$\text{Element b: } \mathbf{K}^b = \begin{array}{c|c} & \begin{array}{cc} \text{B} & \text{C} \end{array} \\ \hline \begin{array}{c} \text{B} \\ \text{C} \end{array} & \begin{bmatrix} 1600 & -1600 \\ -1600 & 1600 \end{bmatrix} \end{array}$$

$$\text{Element c: } \mathbf{K}^c = \begin{array}{c|c} & \begin{array}{cc} \text{C} & \text{D} \end{array} \\ \hline \begin{array}{c} \text{C} \\ \text{D} \end{array} & \begin{bmatrix} 1000 & -1000 \\ -1000 & 1000 \end{bmatrix} \end{array}$$

Assembling these matrices to form the structure stiffness matrix:

$$\mathbf{K} = \begin{array}{c|cccc} & \begin{array}{cccc} \text{A} & \text{B} & \text{C} & \text{D} \end{array} \\ \hline \begin{array}{c} \text{A} \\ \text{B} \\ \text{C} \\ \text{D} \end{array} & \begin{bmatrix} \text{---}2400\text{---} & \text{---}2400\text{---} & & \\ -2400 & 2400 & -1600 & \\ & +1600 & 1600 & -1000 \\ & -1600 & +1000 & -1000 \\ \text{---} & \text{---}1000\text{---} & \text{---}1000\text{---} & \end{bmatrix} & \begin{array}{c} \text{A} \\ \text{B} \\ \text{C} \\ \text{D} \end{array} \end{array}$$

The boundary conditions are $U_A = U_D = 0$. Hence these two rows and columns of $\mathbf{K}$ can be deleted as shown by the dashed lines in the above matrix. Also, the corresponding forces and displacements are excluded from the equilibrium equations, which are then

$$\begin{Bmatrix} 4 \\ 18 \end{Bmatrix} = \begin{bmatrix} 4000 & -1600 \\ -1600 & 2600 \end{bmatrix} \begin{Bmatrix} U_B \\ U_C \end{Bmatrix}$$

The solution to these equations is $U_B = 0.005$ m and $U_C = 0.010$ m. Knowing the displacements, we can now solve for each of the unknown reaction forces by using the equilibrium equation corresponding to that force. These are the same equations which were deleted from the original system of equations. They are

$$F_A = 2400U_A - 2400U_B + 0U_C + 0U_D$$

and

$$F_D = 0U_A + 0U_B - 1000U_C + 1000U_D$$

Substituting the displacements gives $F_A = -12$ kN and $F_D = -10$ kN. Knowing all of the forces permits an equilibrium check to be made, so as to verify the solution (not an essential step, but highly recommended):

$$F_A + F_B + F_C + F_D = -12 + 4 + 18 - 10 = 0$$

5.2 PIN-JOINTED FRAMES

Transformation of Coordinates

At the beginning of this chapter it was shown that a pin-ended bar is equivalent to a spring. Its law of internal force-deformation is

$$\mathcal{F} = \frac{AE}{L} d$$

where $\mathcal{F}$ is the tension or compression in the bar and d is the change in length of the bar. The physical stiffness is AE/L and so from (5.1.5b) the relationship between nodal forces acting on the element and nodal displacements of the element is

$$\begin{Bmatrix} f_1 \\ f_2 \end{Bmatrix} = \frac{AE}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} \quad \text{or} \quad \mathbf{f} = \mathbf{k}^e \mathbf{u} \quad (5.2.1)$$

which establishes $\mathbf{k}^e$, the element stiffness matrix for this element. In this equation the forces and displacements are always axial regardless of how the bar may be orientated in the structure. That is, (5.2.1) is ex-

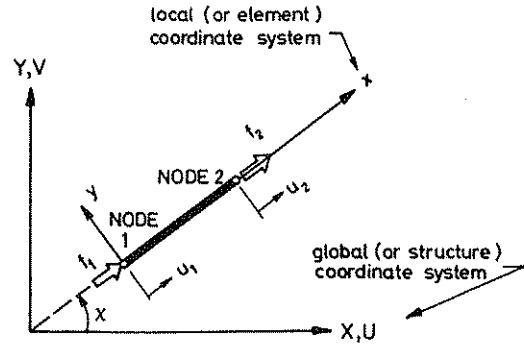


Figure 5.9 Coordinate systems.

pressed in "local" or "element" coordinates (x, y) in which the x -axis is always aligned with the element as shown in Fig. 5.9. In this text all quantities expressed in element coordinates are denoted by a lower case letter ($\mathbf{f}$, $\mathbf{k}$, $\mathbf{u}$, etc.).

In structural frameworks the members occur at various angles to one another and it is necessary to allow for this. In particular, in order to assemble the structure stiffness matrix it is first necessary to express the stiffness matrix of each element of the structure, not in terms of its own element coordinates, but in terms of a "global" or "structure" coordinate system, that is, a coordinate system that is used as the basic reference system for the structure as a whole.

In Fig. 5.9 a pin-ended bar element is inclined at an angle χ to the global system, where χ is positive when measured counterclockwise from the global X -axis to the element x -axis. Axes x and y are the local or element coordinates and X and Y are the structure coordinates. The respective displacements are u and v and U and V , and the respective forces are f_x , f_y , and F_X , and F_Y . Since an axial displacement u of an element generally possesses both a U and V component in the structure coordinate system, it is necessary to expand (5.2.1)

$$\begin{Bmatrix} f_{x1} \\ f_{y1} \\ f_{x2} \\ f_{y2} \end{Bmatrix} = \frac{AE}{L} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} \quad (5.2.2)$$

or

$$\mathbf{f} = \mathbf{k}^e \boldsymbol{\delta}$$

in which the symbols $\mathbf{f}$ and $\boldsymbol{\delta}$ represent the vectors of nodal forces and nodal displacements. That is

$$\mathbf{f} = \begin{Bmatrix} f_{x1} \\ f_{y1} \\ f_{x2} \\ f_{y2} \end{Bmatrix} \quad \text{and} \quad \boldsymbol{\delta} = \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} \quad (5.2.3)$$

Since a pin-ended bar can only carry an axial load, $f_{y1} = f_{y2} = 0$. It may be seen from Fig. 5.10 that the local and global system of forces at node 1 are related by the expressions

$$\begin{aligned} f_{x1} &= F_{x1} \cos \chi + F_{y1} \sin \chi \\ f_{y1} &= -F_{x1} \sin \chi + F_{y1} \cos \chi \end{aligned}$$

and similar expressions apply for node 2. Hence if we define

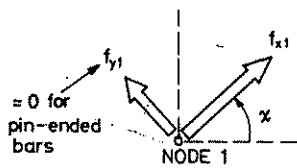
$$\mu = \sin \chi; \quad \lambda = \cos \chi \quad (5.2.4)$$

then the full relationship between the two systems of forces is

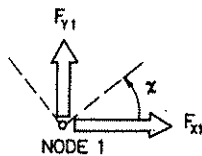
$$\begin{Bmatrix} f_{x1} \\ f_{y1} \\ f_{x2} \\ f_{y2} \end{Bmatrix} = \begin{bmatrix} \lambda & \mu & 0 & 0 \\ -\mu & \lambda & 0 & 0 \\ 0 & 0 & \lambda & \mu \\ 0 & 0 & -\mu & \lambda \end{bmatrix} \begin{Bmatrix} F_{x1} \\ F_{y1} \\ F_{x2} \\ F_{y2} \end{Bmatrix} \quad (5.2.5)$$

or $\mathbf{f} = \mathbf{T}\mathbf{F}$

where $\mathbf{T}$ is called the transformation matrix. It may be



(a) ELEMENT NODAL FORCES AT NODE 1
EXPRESSED IN ELEMENT COORDINATES.



(b) ELEMENT NODAL FORCES AT NODE 1
EXPRESSED IN STRUCTURE COORDINATES

Figure 5.10 Transformation of nodal forces.

shown that this transformation matrix has the very useful property that its inverse is equal to its transpose, that is,

$$\mathbf{T}^{-1} = \mathbf{T}^T$$

For orthogonal systems the transformation between local and global displacements is the same as that between the two sets of forces, namely

$$\boldsymbol{\delta} = \mathbf{T}\mathbf{\Delta} \quad (5.2.6)$$

where

$$\mathbf{\Delta} = \begin{Bmatrix} U_1 \\ V_1 \\ U_2 \\ V_2 \end{Bmatrix}$$

A general transformation involves rotations of three axes x, y, z relative to the structure axes X, Y, Z . The x -axis is at angles θ_{xX}, θ_{xY} , and θ_{xZ} measured from the axes X, Y , and Z as shown in Fig. 5.11 and the cosines of these three angles are the direction cosines of x with respect to X, Y , and Z . They are denoted by $\lambda_{11}, \lambda_{12}$, and λ_{13} . Similarly, the direction cosines of y and z are $\lambda_{21}, \lambda_{22}, \lambda_{23}$ and $\lambda_{31}, \lambda_{32}, \lambda_{33}$ respectively. Thus, the nodal transformation matrix $\mathbf{\Lambda}$ for transforming forces or deflections at each node from structure coordinates to element coordinates is

$$\mathbf{\Lambda} = \begin{bmatrix} \cos \theta_{xX} & \cos \theta_{xY} & \cos \theta_{xZ} \\ \cos \theta_{yX} & \cos \theta_{yY} & \cos \theta_{yZ} \\ \cos \theta_{zX} & \cos \theta_{zY} & \cos \theta_{zZ} \end{bmatrix} \quad (5.2.7)$$

which would be used as follows

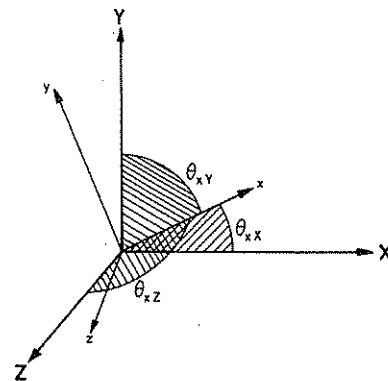


Figure 5.11 General transformation of axes.

$$\mathbf{f}_i = \lambda \mathbf{F}_i \quad \text{and} \quad \delta_i = \lambda \Delta_i$$

where i denotes the node number.

As with $\mathbf{T}$ we have $\lambda^{-1} = \lambda^T$, and therefore the reverse transformations are

$$\mathbf{F}_i = \lambda^T \mathbf{f}_i \quad \text{and} \quad \Delta_i = \lambda^T \delta_i$$

The transformation matrix λ applies at each node and hence for a pin-jointed bar element the full transformation matrix is

$$\mathbf{T} = \begin{bmatrix} \lambda & \mathbf{O} \\ \mathbf{O} & \lambda \end{bmatrix}$$

If the bar element lies in the x, y plane in element coordinates, and in the X, Y plane in structure coordinates, then λ reduces to the simple 2×2 matrix given in (5.2.5)

$$\begin{aligned} \lambda &= \begin{bmatrix} \cos \theta_{xx} & \cos \theta_{xy} \\ \cos \theta_{yx} & \cos \theta_{yy} \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta_{xx} & \cos \left(\frac{\pi}{2} - \theta_{xx} \right) \\ \cos \left(\frac{\pi}{2} + \theta_{xx} \right) & \cos \theta_{xx} \end{bmatrix} \\ &= \begin{bmatrix} \cos \chi & \sin \chi \\ -\sin \chi & \cos \chi \end{bmatrix} \\ &= \begin{bmatrix} \lambda & \mu \\ -\mu & \lambda \end{bmatrix} \end{aligned}$$

Using the transformation matrix $\mathbf{T}$ we can derive an expression for the element stiffness matrix in global coordinates, $\mathbf{K}^e$. The basic force-displacement relationship for the element is given by (5.2.2). If we substitute (5.2.5) on the left-hand side and (5.2.6) for δ on the right-hand side we obtain

$$\mathbf{T}\mathbf{F} = \mathbf{k}^e \mathbf{T}\Delta$$

Premultiplying both sides by $\mathbf{T}^{-1}$ and using the fact that $\mathbf{T}^{-1} = \mathbf{T}^T$ gives

$$\mathbf{F} = \mathbf{T}^T \mathbf{k}^e \mathbf{T} \Delta = \mathbf{K}^e \Delta$$

and therefore the element stiffness matrix in local coordinates $\mathbf{k}^e$ is transformed to the element stiffness matrix in global coordinates $\mathbf{K}^e$ by the compound

transformation

$$\mathbf{K}^e = \mathbf{T}^T \mathbf{k}^e \mathbf{T} \quad (5.2.8)$$

The result of this double multiplication is

$$\mathbf{K}^e = \frac{AE}{L} \begin{bmatrix} \lambda^2 & \lambda\mu & -\lambda^2 & -\lambda\mu \\ \lambda\mu & \mu^2 & -\lambda\mu & -\mu^2 \\ -\lambda^2 & -\lambda\mu & \lambda^2 & \lambda\mu \\ -\lambda\mu & -\mu^2 & \lambda\mu & \mu^2 \end{bmatrix} \quad (5.2.9)$$

It may be seen that $\mathbf{K}^e$ is a symmetric matrix consisting of four sub-matrices κ^e

$$\mathbf{K}^e = \begin{bmatrix} \kappa^e & -\kappa^e \\ -\kappa^e & \kappa^e \end{bmatrix}$$

where

$$\kappa^e = \frac{AE}{L} \begin{bmatrix} \lambda^2 & \lambda\mu \\ \lambda\mu & \mu^2 \end{bmatrix}$$

Unless the elements are all in-line the transformation of the element stiffness matrix from $\mathbf{k}^e$ to $\mathbf{K}^e$ of (5.2.9) must always be performed before the structure stiffness matrix $\mathbf{K}$ can be assembled. Since the arrangement of elements differs for each structure an element stiffness matrix, if it is to be of general use, must first be expressed in terms of its own local coordinates. Also, since the various elements in a structure generally have different sizes and orientations the transformation to structure coordinates must be performed element by element. Hence in the principal steps of matrix stiffness analysis given in Table 5.1, step 2 consists of the transformation of each element stiffness matrix from $\mathbf{k}^e$ to $\mathbf{K}^e$, that is, from local to global coordinates. For all standard elements a general expression for $\mathbf{k}^e$ will be available and so the first part of step 2—determining $\mathbf{k}^e$ —is merely a matter of substituting that element's dimensions and physical properties.

After assembling and solving the complete system of equations (steps 3 to 5 in Table 5.1) the next step in a structural analysis is the calculation of the internal forces in each element. $\mathcal{F} = sd$, where s is the element's material stiffness and d is the deformation. For this it is necessary to convert back to element coordinates in order to determine the element deformation d . For a pin-ended bar the material stiffness is the axial stiffness AE/L . Hence (5.1.10) becomes

$$\mathcal{F}_e = \left(\frac{AE}{L} \right)_e (u_2 - u_1)_e \quad (5.2.10)$$

where u_1 and u_2 are the nodal displacements in element coordinates for element e . Thus before transforming the nodal displacements of the structure from structure coordinates to element coordinates it is first necessary to allocate these displacements to the appropriate ends (nodes) of each element. In a complicated structure this would be done with the aid of the location vectors. After this the transformation from Δ to δ can be performed (separately for each element) and the internal forces can be evaluated. If desired, the transformation can be incorporated into the expression for internal force so that the latter is expressed directly in terms of the structure displacements. In expanded form (5.2.6) is

$$\begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix}_e = \begin{bmatrix} \lambda & \mu & 0 & 0 \\ -\mu & \lambda & 0 & 0 \\ 0 & 0 & \lambda & \mu \\ 0 & 0 & -\mu & \lambda \end{bmatrix} \begin{Bmatrix} U_1 \\ V_1 \\ U_2 \\ V_2 \end{Bmatrix}$$

and therefore

$$\begin{aligned} u_2 - u_1 &= (\lambda U_2 + \mu V_2) - (\lambda U_1 + \mu V_1) \\ &= [\lambda \quad \mu] \begin{Bmatrix} U_2 - U_1 \\ V_2 - V_1 \end{Bmatrix} \end{aligned}$$

Substituting in (5.2.10) gives

$$\mathcal{F}_e = \left(\frac{AE}{L} \right)_e [\lambda \quad \mu]_e \begin{Bmatrix} U_2 - U_1 \\ V_2 - V_1 \end{Bmatrix} \quad (5.2.11)$$

Degrees of Freedom:

Demonstration of Complete Method

The complete process will now be demonstrated for the simple three-bar pin-jointed frame illustrated in Fig. 5.12. The frame is supported at A and C and all members have the same cross-sectional area A and Young's modulus E . The frame carries vertical and horizontal loads as shown in the figure. Since there are three nodes and two independent displacements U and V at each node, the vectors of applied loads and displacements each contain six terms as follows:

$$\mathbf{F} = \begin{Bmatrix} F_{XA} \\ F_{YA} \\ F_{XB} \\ F_{YB} \\ F_{XC} \\ F_{YC} \end{Bmatrix} \quad \text{and} \quad \Delta = \begin{Bmatrix} U_A = 0 \\ V_A = 0 \\ U_B \\ V_B \\ U_C = 0 \\ V_C = 0 \end{Bmatrix}$$

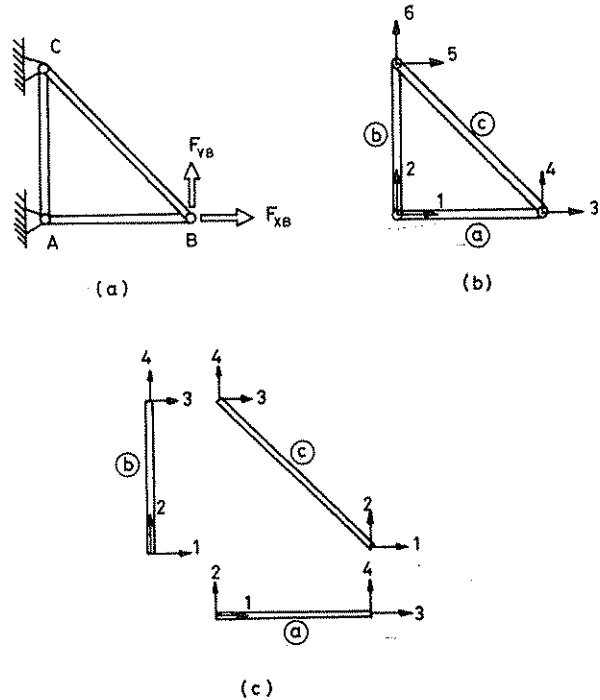


Figure 5.12 Three-member pin-jointed frame. (a) Structure; (b) structure degrees of freedom; and (c) element degrees of freedom.

At this point it is appropriate to introduce the notion of "degrees of freedom." The degrees of freedom of a structure, or of an element, are basically the same as the nodal displacements, but they are slightly more general in concept. Specifically, the degrees of freedom of a structure or element are all of the independent nodal parameters that are used to describe all of the particular ways or modes in which the structure or element can deflect. There are two reasons for introducing this new term:

1. To allow for additional and more general types of "displacement." The most common example is a degree of freedom which is a derivative of an ordinary displacement. For example, in the case of a beam it could be the slope dv/dx or the curvature d^2v/dx^2 at the end (or node) of the beam; these additional "displacements" are derivatives of the beam's ordinary lateral displacement v . The use of this form of displacement will be discussed in the next section.
2. To provide a suitable terminology for the continuous sequential numbering of all of the displacements in a structure (or element), as an alternative to the "two-level" system used thus far which specifies first the node number and then the particular nodal force or displacement at that node. The sequential numbering system based on degrees of freedom is more in harmony with standard matrix notation and is

also more suitable for the practical computer implementation of frame analysis and finite element analysis. Nevertheless, the two-level node numbering is often more suitable for explanatory purposes and we shall continue to make frequent use of it.

The degree of freedom approach is illustrated in Fig. 5.12b which shows that the structure degrees of freedom are numbered sequentially within each node, moving from one node to another. Within each node the numbering should follow some systematic and unchanging sequence (e.g., the X -direction first, then the Y -direction) whereas the nodes may be taken in any sequence. However, we shall see in the next section that the amount of computation can be reduced by a careful choice of node sequence.

Using the numbering system based on structure degrees of freedom, the vectors of applied loads and structure displacements become

$$\mathbf{F} = \begin{Bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \\ F_6 \end{Bmatrix} \quad \text{and} \quad \Delta = \begin{Bmatrix} \Delta_1 \\ \Delta_2 \\ \Delta_3 \\ \Delta_4 \\ \Delta_5 \\ \Delta_6 \end{Bmatrix}$$

The expression for $\mathbf{K}^e$ for a typical element is given by (5.2.9) and to evaluate this the direction cosines λ and μ for each element must be determined. Recalling that χ is measured counter-clockwise from the structure X -axis to the element x -axis, these values are

Member	χ	λ	μ	λ^2	μ^2	$\lambda\mu$
a	0	1	0	1	0	0
b	90	0	1	0	1	0
c	135°	-.707	.707	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$

There must be consistency between the choice of χ and the orientation of the element; i.e. the positioning of element nodes 1 and 2. The element χ axis runs from node 1 to node 2, and therefore χ must be defined at node 1. Thus for member c, if element node 1 is placed at structure node B, then $\chi = 135^\circ$, measured counter-clockwise at node B. If element node 1 is placed at structure node C, then $\chi = 315^\circ$ (or -45°).

Once the structure degrees of freedom have been defined the location vectors can be determined by inspection. Sometimes a diagram such as Fig. 5.12c is of assistance. For each element, one proceeds sequentially through the element degrees of freedom, expressed in structure coordinates, and records the corresponding structure degree of freedom. In a hand solution it is helpful to write the location vector above

and to the right of the transformed stiffness matrix for that element. For the three elements of the sample problem the $\mathbf{K}^e$ matrices and the location vectors are as follows:

Element a:

1	2	3	4
---	---	---	---

$$\mathbf{K}^a = \frac{AE}{L} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{c|c} \hline 1 \\ 2 \\ 3 \\ 4 \\ \hline \end{array}$$

Element b:

1	2	5	6
---	---	---	---

$$\mathbf{K}^b = \frac{AE}{L} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \begin{array}{c|c} \hline 1 \\ 2 \\ 5 \\ 6 \\ \hline \end{array}$$

Element c:

3	4	5	6
---	---	---	---

$$\mathbf{K}^c = \frac{AE}{2\sqrt{2}L} \begin{bmatrix} 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{array}{c|c} \hline 3 \\ 4 \\ 5 \\ 6 \\ \hline \end{array}$$

These matrices are each "broken up" and their terms are assembled together to form the stiffness matrix for the structure. However, before we can do so it is necessary to redefine the assembly process in more general terms. In the initial presentation, in Section 5.1, the example contained only one displacement at each node and so it was possible to use node numbers to identify the rows and columns of the stiffness matrices—both element and structure. It has now been shown that practical problems have multiple displacements (degrees of freedom) at each node, and therefore it is the degree of freedom number which defines the rows and columns of the stiffness matrices—both element and structure. Hence the rule given earlier for performing the assembly process is easily generalized by substituting the term "degree of freedom" in place of "node number." Also, in the earlier example the elements were all in-line and so there was no need to transform $\mathbf{k}^e$ to $\mathbf{K}^e$. Hence to obtain the general form of the rule it is also necessary to replace $\mathbf{k}^e$ by $\mathbf{K}^e$. The result is the following general rule for assembling the structure stiffness matrix $\mathbf{K}$:

For each transformed element stiffness matrix in turn, say $\mathbf{K}^e$, each term of $\mathbf{K}^e$, say K_{lm}^e , is added into the K_{IJ} position of $\mathbf{K}$, where I and J are the structure degrees of freedom corresponding to the element degrees of freedom l and m . The correspondence between I and l and between J and m is indicated by the (previously constructed) location vector for that element.

Applying this rule for the present example gives:

$$\mathbf{K} = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & a_{13} & a_{14} & b_{13} & b_{14} \\ a_{21} + b_{12} & a_{22} + b_{22} & a_{23} & a_{24} & b_{23} & b_{24} \\ a_{31} & a_{32} & a_{33} + c_{11} & a_{34} + c_{12} & c_{13} & c_{14} \\ a_{41} & a_{42} & a_{43} + c_{21} & a_{44} + c_{22} & c_{23} & c_{24} \\ b_{31} & b_{32} & c_{31} & c_{32} & b_{33} + c_{33} & b_{34} + c_{34} \\ b_{41} & b_{42} & c_{41} & c_{42} & b_{43} + c_{43} & b_{44} + c_{44} \end{bmatrix}$$

where a_{ij} represents the term in the i th row and the j th column of $\mathbf{K}^a$, and similarly for b_{ij} and c_{ij} . For each term in an element stiffness matrix the numbers in the location vector above and to the right of that term indicate the column and row in $\mathbf{K}$ where it is to be inserted. For example, the term b_{34} of $\mathbf{K}^b$ is inserted into the $\mathbf{K}_{56}$ location of $\mathbf{K}$. Later, when $\mathbf{K}^c$ is being processed, the term c_{34} is also inserted into the $\mathbf{K}_{56}$ location, being added to b_{34} . In terms of the actual numerical values the fully assembled structure stiffness matrix is

$$\mathbf{K} = \frac{AE}{L} \begin{bmatrix} 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & 1.35 & -0.35 & -0.35 & 0.35 \\ 0 & 0 & -0.35 & 0.35 & 0.35 & -0.35 \\ 0 & 0 & -0.35 & 0.35 & 0.35 & -0.35 \\ 0 & -1 & 0.35 & -0.35 & -0.35 & 1.35 \end{bmatrix}$$

where for brevity 0.35 is written in place of $1/[2(2)^{1/2}] = 0.35355$.

The next step is to impose the boundary conditions and to reduce the system of equations accordingly. For this structure both displacements are zero at node A and at node C. These correspond to degrees of freedom 1, 2, 5, and 6, and therefore these four rows and columns of $\mathbf{K}$, and the corresponding terms in $\mathbf{F}$ and Δ , should be deleted. The reduced system is

$$\begin{Bmatrix} F_3 \\ F_4 \end{Bmatrix} = \frac{AE}{L} \begin{bmatrix} 1.35 & -0.35 \\ -0.35 & 0.35 \end{bmatrix} \begin{Bmatrix} \Delta_3 \\ \Delta_4 \end{Bmatrix}$$

For typical structures the numerical values of F_3 and F_4 would now be inserted and the structure displacements would be obtained by a numerical elimination (or "matrix triangularization") scheme, taking full advantage of the symmetry of $\mathbf{K}$. For such a small system as this one the solution may be obtained in terms of general loads F_3 and F_4 by using matrix inversion. The result is

$$\begin{Bmatrix} \Delta_3 \\ \Delta_4 \end{Bmatrix} = \frac{L}{AE} \begin{bmatrix} 1 & 1 \\ 1 & 3.828 \end{bmatrix} \begin{Bmatrix} F_3 \\ F_4 \end{Bmatrix}$$

If desired, the reaction forces at nodes A and C can now be obtained by substituting Δ_3 and Δ_4 into the equations corresponding to the deleted rows of $\mathbf{K}$.

$$\begin{Bmatrix} F_1 \\ F_2 \\ F_5 \\ F_6 \end{Bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 0 \\ -0.35 & 0.35 \\ 0.35 & -0.35 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 3.828 \end{bmatrix} \begin{Bmatrix} F_3 \\ F_4 \end{Bmatrix}$$

$$= \begin{bmatrix} -1 & -1 \\ 0 & 0 \\ 0 & 1 \\ 0 & -1 \end{bmatrix} \begin{Bmatrix} F_3 \\ F_4 \end{Bmatrix} = \begin{Bmatrix} -F_3 - F_4 \\ 0 \\ F_4 \\ -F_4 \end{Bmatrix}$$

The final step is to calculate the internal forces in the elements. In order to do this the structure displacements Δ must first be distributed amongst the appropriate element nodes, using the location vectors. For the sample problem the correspondence between the displacements is summarized in Table 5.2.

TABLE 5.2 DISTRIBUTION OF STRUCTURE DISPLACEMENTS Δ TO ELEMENT NODES

Element	Element Nodal Displacements (in structure coordinates)			
	U_1	V_1	U_2	V_2
a	Δ_1	Δ_2	Δ_3	Δ_4
b	Δ_1	Δ_2	Δ_5	Δ_6
c	Δ_3	Δ_4	Δ_5	Δ_6

With this information (5.2.11) may now be evaluated for each element in turn. For example, for element a:

$$\begin{aligned} \mathcal{F}_a &= \left(\frac{AE}{L} \right)_a [\lambda \quad \mu]_a \begin{Bmatrix} U_2 - U_1 \\ V_2 - V_1 \end{Bmatrix}_a \\ &= \frac{AE}{L} [1 \quad 0] \begin{Bmatrix} \Delta_3 - \Delta_1 \\ \Delta_4 - \Delta_2 \end{Bmatrix} \\ &= \frac{AE}{L} [1 \quad 0] \frac{L}{AE} \begin{Bmatrix} F_3 + F_4 - 0 \\ F_3 + 3.828F_4 - 0 \end{Bmatrix} \\ &= F_3 + F_4 \end{aligned}$$

Similarly, for element b

$$\begin{aligned}\mathcal{F}_b &= \frac{AE}{L} [0 \quad 1] \begin{Bmatrix} \Delta_5 - \Delta_1 \\ \Delta_6 - \Delta_2 \end{Bmatrix} \\ &= \frac{AE}{L} [0 \quad 1] \frac{L}{AE} \begin{Bmatrix} 0 - 0 \\ 0 - 0 \end{Bmatrix} \\ &= 0\end{aligned}$$

and for element c

$$\begin{aligned}\mathcal{F}_c &= \frac{AE}{\sqrt{2}L} [-.707 \quad .707] \begin{Bmatrix} \Delta_5 & \Delta_3 \\ \Delta_6 & \Delta_4 \end{Bmatrix} \\ &= \frac{AE}{2L} [-1 \quad 1] \frac{L}{AE} \begin{Bmatrix} 0 - F_3 - F_4 \\ 0 - F_3 - 3.828F_4 \end{Bmatrix} \\ &= -1.414F_4\end{aligned}$$

Numbering Sequence of Nodes

If two displacements (or degrees of freedom) in the structure, Δ_i and Δ_m , are linked together by a bar element this will give rise to terms in the structure stiffness matrix in locations K_{ii} , K_{mm} , K_{im} and K_{mi} , as shown in Fig. 5.13. If these two degrees of freedom have widely differing numbers then the equilibrium equations corresponding to F_i will have a wide "bandwidth" w_b , as shown in the figure. In the numerical solution of a system of equations the amount of computation depends on the product $N\bar{w}_b^2$, where N is the total number of degrees of freedom and $\bar{w}_b$ is the average bandwidth. For a given structure N is a constant and so for the present discussion we may ignore the effect of N . Therefore in the degree of freedom numbering the order in which the nodes are considered should be such that the average difference in the degree of freedom numbers is as small as possible. The rule for achieving this is simple: the numbering should always proceed across the shortest width of the structure, thus only gradually moving along the length of the structure. This is illustrated in the 13-bar pin-

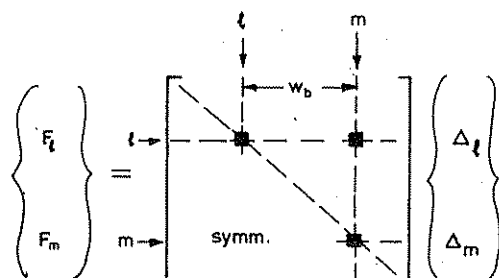


Figure 5.13 Bandwidth of the structural stiffness matrix.

jointed structure of Fig. 5.14. To simplify the discussion we can use node numbers instead of degrees of freedom, because this will not alter the conclusion in any way. It simply means that since there are two degrees of freedom at each node, each square in the stiffness matrix diagram represents four terms. In the first case (Fig. 5.14a) this rule is not followed and $\bar{w}_b$ is 3.9, whereas if the rule is followed (Fig. 5.14b) the value is 2.7. Thus the first node sequence requires $(3.9/2.7)^2 = 2.09$ times as much computation as the second sequence.

5.3 BEAM ELEMENT: RIGID-JOINTED FRAME ANALYSIS

Flexure-Only Beam Element

For several obvious reasons the two- and three-dimensional frameworks found in ships are rigid-jointed rather than pin-jointed: the need for watertightness, the use of welding, and the large lateral loads which must be carried. The basic element for a rigid-jointed frame is the beam element shown in Fig. 5.15. The beam is loaded by forces and moments at each node and is assumed to be of uniform flexural rigidity EI .

By definition, the nodes of a beam element are the points where the external forces and moments act, and these are assumed to be in equilibrium with the internal forces which act along the beam's neutral axis. Therefore, unless special provision is made for nodal eccentricity, the nodes of a beam element are located at the centroid of the beam's cross section.

The element stiffness matrix k^e may again be obtained by following the same procedure as for the bar element, that is, by imposing each of the nodal displacements in turn, determining the nodal forces set up, and then superimposing the results obtained from each individual case. Alternatively, since the element is a prismatic beam, we may make use of beam theory to obtain the required relationships between the nodal forces and displacements. Specifically, these relationships may be obtained from the well-known slope-deflection equations of beam theory. For the element shown in Fig. 5.15 the slope-deflection equations are

$$\begin{aligned}M_1 &= \frac{6EI}{L^2} v_1 + \frac{4EI}{L} \theta_1 - \frac{6EI}{L^2} v_2 + \frac{2EI}{L} \theta_2 \\ M_2 &= \frac{6EI}{L^2} v_1 + \frac{2EI}{L} \theta_1 - \frac{6EI}{L^2} v_2 + \frac{4EI}{L} \theta_2\end{aligned}$$

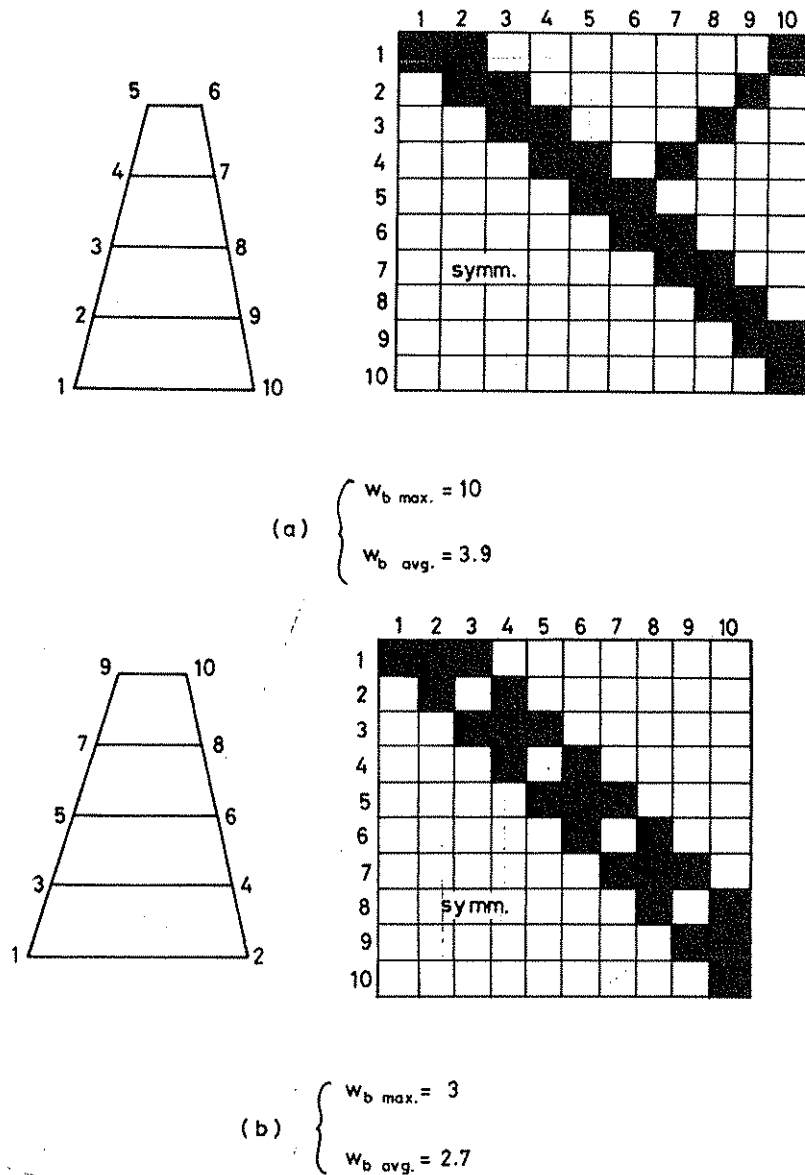


Figure 5.14 Effect of structure node numbering on bandwidth.

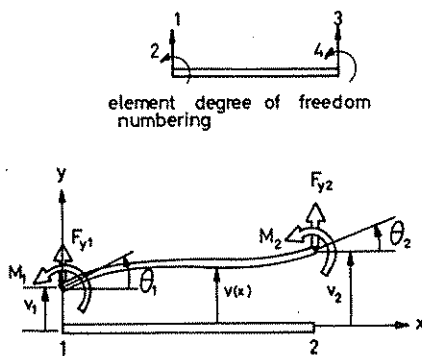


Figure 5.15 Flexure-only beam element.

From vertical force equilibrium

$$F_{y1} = -F_{y2} = \frac{M_1 + M_2}{L}$$

$$= \frac{12EI}{L^3} v_1 + \frac{6EI}{L^2} \theta_1 - \frac{12EI}{L^3} v_2 + \frac{6EI}{L^2} \theta_2$$

In matrix notation these equations are

$$\begin{Bmatrix} F_{y1} \\ M_1 \\ F_{y2} \\ M_2 \end{Bmatrix} = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix} \begin{Bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{Bmatrix} \quad (5.3.1)$$

or $\mathbf{f} = \mathbf{k}^e \delta$

which defines the nodal force vector $\mathbf{f}$, the nodal displacement vector δ , and the stiffness matrix $\mathbf{k}^e$ for a beam element in which only the flexural response is considered. It is important to note that in this element axial forces and axial deformation are deliberately ignored. Hence this element can only be used for applications in which the loads are carried purely by beam bending, such as in laterally loaded flat grillages, or in which the axial displacements and flexural displacements are completely independent, such as in orthogonal frameworks, providing also that the structure does not undergo "sway," that is, translation of joints so as to cause axial rigid body motion of any element. Hence this element cannot be used for trusses or for other frame structures in which a given load is carried partly by axial stiffness and partly by bending stiffness, such as a pitched grillage (e.g., a ship's bottom that has some deadrise). In fact, about the only types of structure for which this element can be used are multispans beams and flat grillages. The latter are particularly common in ships, and this application is discussed in the next section. The element is introduced here at this point because it allows a simpler explanation of some of the principles and techniques of frame analysis.

The nodal displacements given above include the rotation θ at the end of the beam. It should be noted that in small-deflection beam theory the quantity that is actually involved is the *slope* of the beam, that is, the derivative $v' = dv/dx$; for small deflections this is equivalent to the angle of rotation and the latter term is usually chosen because a rotation is easier to visualize as a form of displacement. However, for basic derivations the derivative term is preferable because it shows that within the element the deflected state of the beam is described completely by the general displacement function $v(x)$; the slope at any point—including the ends—is simply the derivative of $v(x)$ at that point. Thus the four nodal "displacements" are in reality the nodal values of $v(x)$ and of $v'(x)$, and this illustrates why the name "degree of freedom" is often used instead of "displacement."

For an arbitrarily orientated beam element it is again necessary to expand $\mathbf{k}^e$ to allow for the element displacements v (which are always normal to the beam) to be transformed into U and V displacements in the global coordinate system. The nodal bending moments are unaffected. Consequently $\mathbf{T}$ takes the form

$$\mathbf{T} = \begin{bmatrix} \lambda & \mu & 0 & 0 & 0 & 0 \\ -\mu & \lambda & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda & \mu & 0 \\ 0 & 0 & 0 & -\mu & \lambda & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (5.3.2)$$

As shown in (5.2.8) the transformation of the element stiffness matrix from local to global coordinates is $\mathbf{T}^T \mathbf{k}^e \mathbf{T}$. For the beam element the result is

$$\mathbf{K}^e = \frac{EI}{L^3} \begin{bmatrix} \kappa_1 & \kappa_2^T \\ \kappa_2 & \kappa_3 \end{bmatrix} \quad (5.3.3)$$

where

$$\kappa_1 = \begin{bmatrix} 12\mu^2 & \text{symm.} \\ -12\lambda\mu & 12\lambda^2 \\ -6\mu L & 6\lambda L & 4L^2 \end{bmatrix}$$

$$\kappa_3 = \begin{bmatrix} 12\mu^2 & \text{symm.} \\ -12\lambda\mu & 12\lambda^2 \\ 6\mu L & -6\lambda L & 4L^2 \end{bmatrix}$$

and

$$\kappa_2 = \begin{bmatrix} -12\mu^2 & 12\lambda\mu & 6\mu L \\ 12\lambda\mu & -12\lambda^2 & -6\lambda L \\ -6\mu L & 6\lambda L & 2L^2 \end{bmatrix}$$

From this point on, the use of this element for the structural analysis of a rigid-jointed frame would proceed along the remaining steps of Table 5.1: the overall stiffness matrix $\mathbf{K}$ is assembled, the support conditions are applied, and the resulting set of equations solved. However, before proceeding further in regard to the application of the beam element, we shall use it as an example to demonstrate the general method for defining an element and deriving its stiffness matrix.

General Method for Deriving An Element Stiffness Matrix

In presenting the general method we choose the flexure-only beam element as the example because this

element is quite simple and yet is sufficiently general to illustrate all of the basic steps in the definition of an element and the derivation of its stiffness matrix. In Chapter 7 it will be shown that in finite element theory this same method is used to derive the element stiffness matrix for a wide variety of two- and three-dimensional finite elements. Hence this method constitutes one of the basic foundations of matrix structural analysis. In terms of the seven steps of matrix structural analysis given in Table 5.1, this method provides the material which is essential for step 2, that is, the complete definition of the element properties, and especially the element stiffness matrix k^e . The method consists of five steps, and in order to distinguish them from the seven basic steps of matrix structural analysis we shall identify them by means of Roman numerals.

STEP I: SELECT SUITABLE DISPLACEMENT FUNCTION

The first step is to specify a displacement function which uniquely defines the state of displacement at all points within the element in terms of the nodal degrees of freedom. For a beam the displacement within the element is $v(x)$. This displacement can conveniently be represented by a polynomial expression and since the aim is to express $v(x)$ in terms of the nodal degrees of freedom δ , the assumed polynomial must contain one unknown coefficient for each degree of freedom possessed by the element. As defined previously the beam element possesses four degrees of freedom

$$\delta = \begin{Bmatrix} v_1 \\ v'_1 \\ v_2 \\ v'_2 \end{Bmatrix} \quad (5.3.4)$$

and therefore we choose a cubic polynomial to represent $v(x)$, since this has four coefficients.

$$v(x) = C_1 + C_2x + C_3x^2 + C_4x^3 \quad (5.3.5)$$

In matrix form this is

$$v(x) = \mathbf{H}(x)\mathbf{C} \quad (5.3.6)$$

where $\mathbf{H}(x) = [1 \quad x \quad x^2 \quad x^3]$

and $\mathbf{C}^T = [C_1 \quad C_2 \quad C_3 \quad C_4]$

Equation (5.3.4) defines the vector of nodal displacements δ . We now construct the *generalized dis-*

placement vector $\delta(x)$ which expresses the values of the displacements $v(x)$ and $v'(x)$ at any point within the element, in terms of the four (as yet undetermined) coefficients. Differentiating (5.3.5) gives

$$v'(x) = C_2 + 2C_3x + 3C_4x^2$$

and by writing this in matrix form and combining it with (5.3.6) we obtain

$$\delta(x) = \begin{Bmatrix} v(x) \\ v'(x) \end{Bmatrix} = \begin{bmatrix} 1 & x & x^2 & x^3 \\ 0 & 1 & 2x & 3x^2 \end{bmatrix} \begin{Bmatrix} C_1 \\ C_2 \\ C_3 \\ C_4 \end{Bmatrix} \quad (5.3.7)$$

STEP II: RELATE GENERAL DISPLACEMENTS WITHIN THE ELEMENT TO ITS NODAL DISPLACEMENTS

The coefficients of the displacement function $\mathbf{C}$ are now expressed in terms of the nodal displacements δ and hence by substituting into (5.3.7) the displacements at any point within the element are obtained in terms of the nodal displacements δ .

Since $\delta(x)$ represents the displacement at any point x the nodal displacements can be obtained from it by simply substituting the appropriate nodal coordinates into (5.3.7).

At node 1, $x = 0$ and therefore

$$v_1 = C_1 \quad \text{and} \quad v'_1 = C_2$$

At node 2, $x = L$ and therefore

$$v_2 = C_1 + C_2L + C_3L^2 + C_4L^3$$

$$v'_2 = C_2 + 2C_3L + 3C_4L^2$$

Writing these in matrix form

$$\begin{Bmatrix} v_1 \\ v'_1 \\ v_2 \\ v'_2 \end{Bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & L & L^2 & L^3 \\ 0 & 1 & 2L & 3L^2 \end{bmatrix} \begin{Bmatrix} C_1 \\ C_2 \\ C_3 \\ C_4 \end{Bmatrix}$$

or $\delta = \mathbf{A}\mathbf{C}$

Since the matrix $\mathbf{A}$ is known, this equation may be solved for the vector of unknown coefficients $\mathbf{C}$

$$\mathbf{C} = \mathbf{A}^{-1} \delta \quad (5.3.8a)$$

and it may be shown that

$$\mathbf{A}^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{-3}{L^2} & \frac{-2}{L} & \frac{3}{L^2} & \frac{-1}{L} \\ \frac{2}{L^3} & \frac{1}{L^2} & \frac{-2}{L^3} & \frac{1}{L^2} \end{bmatrix} \quad (5.3.8b)$$

For future use we note here that (5.3.6) and (5.3.8) can be combined to obtain an expression for the displacement within the element, $v(x)$, in terms of the nodal displacements δ . Combining these two equations gives

$$v(x) = \mathbf{H}(x)\mathbf{A}^{-1}\delta$$

and the result of the matrix multiplication is

$$v(x) = \mathbf{N}(x)\delta \quad (5.3.9a)$$

in which $\mathbf{N}(x)$ is known as the *shape function* and is given by

$$\mathbf{N}(x) = \begin{bmatrix} (1 - 3\xi^2 + 2\xi^3) & x(1 - 2\xi + \xi^2) \\ (3\xi^2 - 2\xi^3) & x(-\xi + \xi^2) \end{bmatrix} \quad (5.3.9b)$$

where $\xi = x/L$.

STEP III: EXPRESS THE INTERNAL DEFORMATION IN TERMS OF THE NODAL DISPLACEMENTS

The definition of "deformation" of an element depends on which particular field of solid mechanics is involved. In plane elasticity the deformation is strain $\varepsilon(x, y)$, whereas for beam bending it is the curvature of the beam. In each case the deformation is some form of derivative of the displacement; in plane elasticity it is a first derivative ($\varepsilon_x = \partial u / \partial x$, etc.) whereas the curvature of a beam is (for small deflections) the second derivative of the beam displacement, $v''(x) = d^2v/dx^2$. Therefore, from (5.3.5)

$$v''(x) = 2C_3 + 6C_4x$$

or in matrix form

$$v''(x) = \begin{bmatrix} 0 & 0 & 2 & 6x \end{bmatrix} \begin{Bmatrix} C_1 \\ C_2 \\ C_3 \\ C_4 \end{Bmatrix}$$

From (5.3.8a)

$$v''(x) = \begin{bmatrix} 0 & 0 & 2 & 6x \end{bmatrix} \mathbf{A}^{-1} \delta$$

and substitution for $\mathbf{A}^{-1}$ followed by matrix multiplication gives

$$v''(x) = \mathbf{B} \delta$$

where

(5.3.10)

$$\mathbf{B} = \begin{bmatrix} \frac{-6}{L^2} + \frac{12x}{L^3} & \frac{-4}{L} + \frac{6x}{L^2} & \frac{6}{L^2} - \frac{12x}{L^3} & \frac{-2}{L} + \frac{6x}{L^2} \end{bmatrix}$$

STEP IV: EXPRESS THE INTERNAL FORCE IN TERMS OF THE NODAL DISPLACEMENTS, USING THE ELEMENT'S LAW OF ELASTIC BEHAVIOR

As with internal deformation, the definition of internal force depends on the class of problem being considered. For problems in two- and three-dimensional elasticity it consists of the components of stress, whereas for beam bending it is the internal bending moment in the beam, $M_z(x)^*$. In this step the goal is to express the internal force in terms of the nodal

*The use of script symbols for internal forces is especially helpful in the case of bending moment, M , because it is necessary to distinguish it from an applied (nodal) moment M ; these two quantities are not the same and they have a different sign convention. The term "bending moment" is used to describe the internal state of the beam, that is, its response to the external load. The deformation response is a curvature and so the sign of the internal force response (i.e., bending moment M) is determined by the curvature of the beam: $M_z(x)$ is positive if $v''(x)$ is positive. A nodal moment M is an applied load, a certain type of force (a couple) that acts at a particular location. It may be either an external load acting on the structure or an element nodal moment acting on a particular element. Either way, its sign is determined by the coordinate system for forces and deflections (using the right hand rule), either at the structure level or at the element level. Thus, M_z is a moment acting about the z -axis and it is positive if its rotation would give an advance in the $+z$ -direction. Of course, at a node (say element node 1) the internal bending moment $M_z(0)$, or M_{z1} , would be equal in magnitude to the element nodal moment acting at that node, M_{z1} , but the sign could be different, and this could be a source of confusion and error if these two are not clearly distinguished. In small problems which are solved by hand the possibility of confusion and error is perhaps not so serious but with large problems and for computer implementation it is important to distinguish between these two quantities. In addition to using different symbols, in this text the phrase "bending moment" shall be used only for M , and never for M . Other terms, such as (element) nodal moment, generalized nodal force (or load), applied moment, and so on, which do not contain the phrase "bending moment," will be used for M .

displacements. Since a relationship between the internal deformation and the nodal displacements δ is already known, the internal force can be related to the nodal displacements providing that the element's internal force-deformation relationship is known. For elasticity problems this relationship is Hooke's Law. For beam bending it is the moment-curvature relationship

$$M_z(x) = EIv''(x) \quad (5.3.11)$$

Therefore the expression for the internal force in terms of nodal displacements is

$$M_z(x) = EIB \delta$$

From the expression for B in (5.3.10) it can be seen that in the beam element the bending moment varies linearly along the beam.

For any element the expression for internal force is used not only in deriving the element stiffness matrix but also in any structural analysis which makes use of that element, because step 6 of matrix structural analysis is the calculation of the internal forces in the various elements by transforming the nodal displacements back into element coordinates and substituting them into the expression for internal force. From (5.3.10) the expression is

$$M_z(x) = b \begin{bmatrix} -6L + 12x & -4L^2 + 6Lx \\ 6L - 12x & -2L^2 + 6Lx \end{bmatrix} \begin{Bmatrix} v_1 \\ v_1' \\ v_2 \\ v_2' \end{Bmatrix} \quad (5.3.12a)$$

$$\text{in which } b = \frac{EI}{L^3}.$$

In many cases it is desired, or is sufficient, to obtain the internal forces only at the nodal points. In the beam element, for example, the linear distribution of $M_z(x)$ means that it is sufficient to calculate M_z only at the nodes. An expression for these nodal values is easily obtained by simply substituting $x = 0$ and $x = L$ into the foregoing expression. The result is

$$\begin{Bmatrix} M_z(0) \\ M_z(L) \end{Bmatrix} = S \delta \quad (5.3.12b)$$

where

$$S = \frac{EI}{L^3} \begin{bmatrix} -6L & -4L^2 & 6L & -2L^2 \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}$$

STEP V: OBTAIN THE ELEMENT STIFFNESS MATRIX k^e BY RELATING NODAL FORCES TO NODAL DISPLACEMENTS

In this step the internal forces are replaced by statically equivalent nodal forces f and hence the nodal forces are related to the nodal displacements δ , thereby defining the required element stiffness matrix k^e .

The principle of virtual work is used to determine the set of nodal forces that is statically equivalent to the internal forces. The condition of equivalence may be expressed as follows: during any virtual displacement imposed on the element, the total external work done by the nodal forces must equal the total internal work done by the internal forces. While the virtual displacement is being imposed all actual forces, both nodal and internal, are constant. The arbitrary virtual nodal displacements are represented by the symbol δ^* ; that is

$$\delta^* = \begin{Bmatrix} v_1^* \\ v_1'^* \\ v_2^* \\ v_2'^* \end{Bmatrix}$$

We begin with the external virtual work. If the above virtual nodal displacements are imposed on an element in which the actual nodal forces are

$$f = \begin{Bmatrix} f_1 \\ M_1 \\ f_2 \\ M_2 \end{Bmatrix}$$

then the virtual work done externally is

$$\begin{aligned} W_{\text{ext}} &= v_1^* f_1 + v_1'^* M_1 + v_2^* f_2 + v_2'^* M_2 \\ &= \delta^{*T} f \end{aligned}$$

Next we evaluate the internal virtual work. If an arbitrary virtual curvature $v''(x)^*$ is imposed on a beam of length L in which the actual internal bending moment is $M_z(x)$, the internal virtual work is

$$W_{\text{int}} = \int_0^L (v''(x)^*)^T M_z(x) dx$$

If the virtual curvature is imposed indirectly, in terms of virtual nodal displacements δ^* , the corre-

sponding value of v''^* is

$$(v''(x)^*)^T = (\mathbf{B} \boldsymbol{\delta}^*)^T = \boldsymbol{\delta}^{*T} \mathbf{B}^T$$

In matrix algebra, expanding the transpose of a product causes the terms to be reversed. The actual internal bending moment can be expressed in terms of the actual curvature $v''(x)$ by means of (5.3.11) and so the internal work resulting from the virtual nodal displacement is

$$W_{\text{int}} = \int_0^L \boldsymbol{\delta}^{*T} \mathbf{B}^T EI v''(x) dx$$

Finally, from (5.3.10) the actual curvature $v''(x)$ can be expressed in terms of the actual nodal displacement $\boldsymbol{\delta}$. The result is

$$W_{\text{int}} = \int_0^L \boldsymbol{\delta}^{*T} \mathbf{B}^T EI \mathbf{B} \boldsymbol{\delta} dx$$

Equating the internal and external work and setting the arbitrary virtual displacement $\boldsymbol{\delta}^*$ to unity (we can do this because the value is arbitrary) gives

$$\mathbf{f} = \left[\int_0^L \mathbf{B}^T EI \mathbf{B} dx \right] \boldsymbol{\delta}$$

Therefore since this equation relates nodal force $\mathbf{f}$ to nodal displacement $\boldsymbol{\delta}$ the quantity in brackets must be the element stiffness matrix $\mathbf{k}^e$. Substituting for $\mathbf{B}$ and performing the integration produces the same stiffness matrix which was derived earlier using the slope-deflection equations and was presented in (5.3.1).

Ordinary Beam Element

For general frame structures in which the members are not orientated orthogonally the axial stiffness and the flexural stiffness are not independent and the former must be included in the element stiffness matrix. Hence the usual beam element has six degrees of freedom as shown in Fig. 5.16. These are usually taken in the following order.

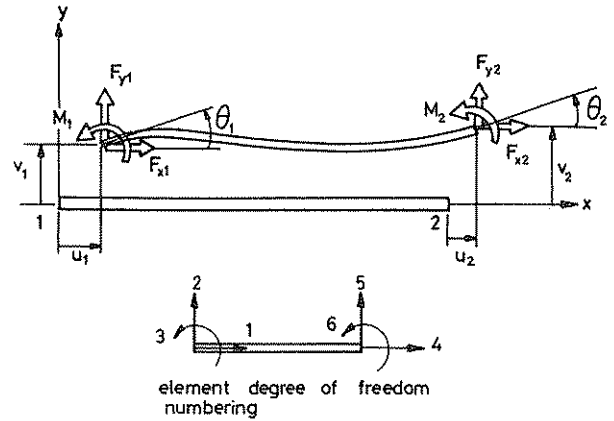


Figure 5.16 Ordinary beam element.

$$\boldsymbol{\delta} = \begin{Bmatrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \end{Bmatrix} \quad (5.3.13)$$

Accordingly, in the 6×6 element stiffness matrix the axial stiffness terms occur in the first and fourth rows and columns, and they are the same as for the pinned bar element. For convenience we define the following axial and flexural parameters

$$\alpha = \frac{AE}{L} \quad \text{and} \quad b = \frac{EI}{L^3} \quad (5.3.14)$$

In terms of α and b the stiffness matrix of the ordinary beam element is

$$\mathbf{k}^e = \begin{bmatrix} \alpha & 0 & 0 & -\alpha & 0 & 0 \\ 0 & 12b & 6bL & 0 & -12b & 6bL \\ 0 & 6bL & 4bL^2 & 0 & -6bL & 2bL^2 \\ -\alpha & 0 & 0 & \alpha & 0 & 0 \\ 0 & -12b & -6bL & 0 & 12b & -6bL \\ 0 & 6bL & 2bL^2 & 0 & -6bL & 4bL^2 \end{bmatrix} \quad (5.3.15)$$

which can be recognized as the "flexure-only" stiffness matrix plus the axial stiffness terms.

As before, the element stiffness matrix in structure coordinates is $\mathbf{T}^T \mathbf{k}^e \mathbf{T}$ with $\mathbf{T}$ defined in (5.3.2). The result is

$$\mathbf{K}^e = \begin{bmatrix} \alpha\lambda^2 + 12b\mu^2 & (\alpha - 12b)\lambda\mu & -6bL\mu & \alpha\mu^2 + 12b\lambda^2 & 6bL\lambda & 4bL^2 \\ -\alpha\lambda^2 - 12b\mu^2 & -(\alpha - 12b)\lambda\mu & -6bL\mu & -\alpha\mu^2 - 12b\lambda^2 & -6bL\lambda & 2bL^2 \\ \alpha\lambda^2 + 12b\mu^2 & (\alpha - 12b)\lambda\mu & -6bL\mu & \alpha\mu^2 + 12b\lambda^2 & 6bL\lambda & 4bL^2 \\ -\alpha\lambda^2 - 12b\mu^2 & -(\alpha - 12b)\lambda\mu & -6bL\mu & -\alpha\mu^2 - 12b\lambda^2 & -6bL\lambda & 2bL^2 \end{bmatrix} \quad (5.3.16)$$

symmetric

The expression for the internal forces is simply a combination of (5.2.10) for the axial force and (5.3.12) for the bending moment. In the former we shall divide by the beam cross-sectional area A to obtain axial stress σ_x . If desired, an expression for the shear force can also be included since the shear force is simply the derivative of the internal bending moment: $\mathcal{Q}(x) = (d/dx)[\mathcal{M}_x(x)]$. From (5.3.10) and (5.3.11)

$$\mathcal{Q}(x) = \frac{EI}{L^3} [12 \quad 6L \quad -12 \quad 6L] \delta \quad (5.3.17) \quad \text{and thus}$$

Thus the shear force is constant along the beam element, as is the axial force. Therefore it is sufficient to have an expression for the forces only at the nodes. This is

$$\begin{Bmatrix} \sigma_{x1} \\ \mathcal{Q}_1 \\ \mathcal{M}_{x1} \\ \sigma_{x2} \\ \mathcal{Q}_2 \\ \mathcal{M}_{x2} \end{Bmatrix} = \begin{bmatrix} -\frac{E}{L} & 0 & 0 & \frac{E}{L} & 0 & 0 \\ 0 & 12b & 6bL & 0 & -12b & 6bL \\ 0 & -6bL & -4bL^2 & 0 & 6bL & -2bL^2 \\ -\frac{E}{L} & 0 & 0 & \frac{E}{L} & 0 & 0 \\ 0 & 12b & 6bL & 0 & -12b & 6bL \\ 0 & 6bL & 2bL^2 & 0 & -6bL & 4bL^2 \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \end{Bmatrix} \quad (5.3.18)$$

Note that although these are nodal values the script form is used for $\mathcal{M}$ and $\mathcal{Q}$ in order to indicate that they are internal forces.

Restraints and Specified Displacements

Earlier, when discussing boundary conditions and restraints, it was shown that the imposition of zero displacement at a node can be achieved by simply deleting that row and column from the structure equilibrium equations. This is quite suitable if the equations are being solved by hand (any reduction in the number of equations is then most valuable) but it is not so suitable for a computer solution. Deletion of rows and columns from $\mathbf{K}$ would involve repacking it within the computer, and this is time-consuming. A more efficient technique is to replace the diagonal terms of $\mathbf{K}$ corresponding to the zero displacement by unity and replace the rest of the terms in the corresponding row and column by zeros.

A second method is available which requires only one multiplication for each degree of freedom that is to

be eliminated. In this method each diagonal term of $\mathbf{K}$ that corresponds to a zero displacement is multiplied by a large number such as 10^{25} . For example, suppose the displacement which is degree of freedom number I is to be zero. After the multiplication the equilibrium equation corresponding to that degree of freedom is

$$F_I = K_{I1}\Delta_1 + K_{I2}\Delta_2 + \cdots + 10^{25}K_{II}\Delta_I + \cdots + K_{IN}\Delta_N$$

$$\Delta_I = \frac{F_I - (K_{I1}\Delta_1 + K_{I2}\Delta_2 + \cdots + K_{IN}\Delta_N)}{10^{25}K_{II}}$$

Since the diagonal term is now so much greater than the off-diagonal terms, the value of the displacement Δ_I is very close to zero.

If a displacement Δ_I is required to have some specified nonzero value d then this effect too may be allowed for using either of these methods, together with an adjustment of the load vector. If the first method is used, then after placing the 1 and the zeros in $\mathbf{K}$ the F_I term in the force vector is replaced by the value of the prescribed displacement d . Alternatively, using the second method, the F_I term is replaced by the diagonal term multiplied by the prescribed displacement. Thus writing out the equation for row I

$$10^{25}K_{II}d = K_{I1}\Delta_1 + K_{I2}\Delta_2 + \cdots + 10^{25}K_{II}\Delta_I + \cdots + K_{IN}\Delta_N$$

Consequently

$$\Delta_I = \frac{10^{25}K_{II}d - (K_{I1}\Delta_1 + K_{I2}\Delta_2 + \cdots + K_{IN}\Delta_N)}{10^{25}K_{II}}$$

Since the diagonal term $10^{25}K_{II}$ is so much larger than the off-diagonal terms, then to a good approximation $\Delta_I = d$.

Distributed Loads: Equivalent Nodal Loads

In frame analysis, and also in finite element analysis, all quantities are specified or represented in terms of their values at the nodes. Until now we have assumed that the loads occur only at nodes whereas in most frame structures, and certainly in ships, the loads are distributed loads. The technique for dealing with such loads is to divide the analysis into two parts, a local analysis which deals only with the internodal loading and response of the loaded beam, and an overall analysis which deals with the response of the overall structure. A separate solution is obtained for each, and then the solutions are superimposed. For this purpose the distributed load is divided into two parts: an internal load which is applied within the beam and which accounts for the internodal bending; and an external load which is applied at the boundaries of the loaded beam (i.e., at the nodes) and which acts on the entire structure. Since it is a nodal load the usual frame analysis method may be used for the solution.* Obviously each load must give an accurate solution within its own field, and the two loads taken together must match the original load. In the case of beam bending this is achieved by defining the internal load as the original distributed load plus equilibrium end forces and moments that correspond to complete fixity at the boundary, that is, zero deflection and zero rotation at the ends (or nodes) of the loaded beam. In this way the internal problem satisfies equilibrium and may be dealt with in isolation from the overall solution. Thus the internal problem is simply the given distributed load acting on a clamped beam, and for standard types of distributed loading the solution is already available. In particular, the bending moments at the ends of the beam are simply the "fixed-end moments" for that loading. Figure 5.17 gives the fixed-end reaction forces and moments for a variety of distributed loads commonly occurring in ship structures.

The external load must satisfy three requirements:

1. It must consist of nodal forces and moments.
2. In regard to its effect on the overall solution it must be statically equivalent to the actual distributed load.
3. The combination of the internal and external loads must be the actual load.

From (3) the external load must be equal and opposite to the clamping forces and moments which were added to the distributed load in the internal problem. Since these clamping forces and moments equilibrated the distributed load they were statically equivalent to it but opposite in sign. Hence the forces and moments in the external problem, being of opposite sign to these clamping forces and moments, will be exactly statically equivalent to the distributed load. Thus in any frame analysis the statically equivalent nodal loads which are used in place of a distributed load are exactly the same, but opposite in sign, as the fixed-end reaction forces and moments corresponding to that load.

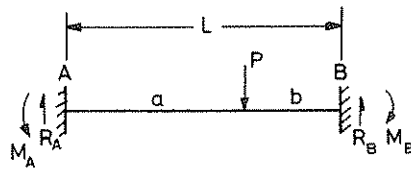
Although the equivalent nodal loads are an adequate approximation of the distributed loads in regard to obtaining the structural (nodal) displacements, they are not sufficient in themselves to obtain the nodal forces and moments. Since these equivalent loads have been added artificially at the nodes, the nodal values of the element internal forces (shear force and bending moment) which are calculated in step 6 of Table 5.1 will be too large by this amount, and the excess must be subtracted. Therefore if an equivalent load has been used for any element, the fixed-end forces and moments must be added to the nodal forces and moments obtained from step 6, regardless of whether the full internal distribution of shear force and bending moment is required or not. This correction constitutes step 7 in the overall procedure of matrix stiffness analysis.

Example

The portal frame of Fig. 5.18 will be used as an example to demonstrate several aspects of matrix stiffness analysis which have been discussed previously:

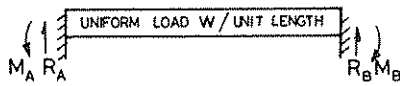
1. The modeling of symmetry in a structure.
2. The use of equivalent nodal loads to represent a distributed load.
3. The relative magnitude of axial and flexural stiffness and the order of accuracy attainable when neglecting the axial degree of freedom in orthogonal frames.

*This technique is used in several branches of continuum mechanics. For example, in fluid mechanics a flow is often represented by the combination of a "near field" flow which accounts for the local aspects (e.g., flow around a body) and a "far field" flow which accounts for the effects in the rest of the field.



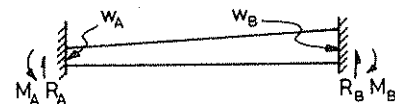
$$R_A = \frac{Pb^2(3a+b)}{L^3} \quad R_B = \frac{Pa^2(a+3b)}{L^3}$$

$$M_A = \frac{Pab^2}{L^2} \quad M_B = -\frac{Pa^2b}{L^2}$$



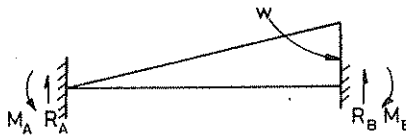
$$R_A = \frac{WL}{2} \quad R_B = \frac{WL}{2}$$

$$M_A = \frac{WL^2}{12} \quad M_B = -\frac{WL^2}{12}$$



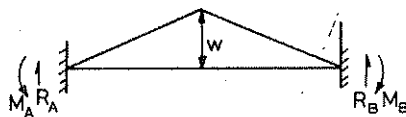
$$R_A = \frac{L}{6}(2w_A + w_B) \quad R_B = \frac{L}{6}(w_A + 2w_B)$$

$$M_A = \frac{L^2}{60}(3w_A + 2w_B) \quad M_B = \frac{L^2}{60}(2w_A + 3w_B)$$



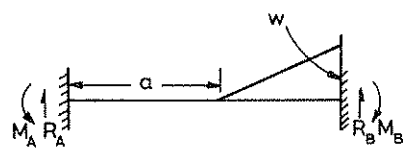
$$R_A = \frac{3}{20}wL \quad R_B = \frac{7}{20}wL$$

$$M_A = \frac{wL^2}{30} \quad M_B = -\frac{wL^2}{20}$$



$$R_A = \frac{wL}{4} \quad R_B = \frac{wL}{4}$$

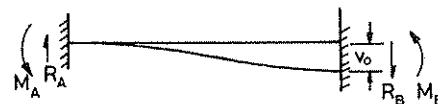
$$M_A = \frac{5wL^2}{96} \quad M_B = -\frac{5wL^2}{96}$$



$$R_A = \frac{w(L-a)^3(3L+2a)}{20L^3} \quad R_B = \frac{w(L-a)}{20L^3} [10L^3 - (L-a)^2(3L+2a)]$$

$$M_A = \frac{w}{60L^2} (L-a)^3 (2L+3a)$$

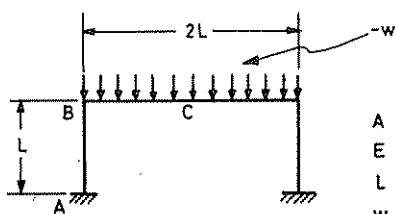
$$M_B = \frac{-w}{60L^2} (L-a)^2 (3L^2 + 4aL + 3a^2)$$



$$R_A = -R_B = \frac{12 EI v_0}{L^3}$$

$$M_A = M_B = \frac{6 EI v_0}{L^2}$$

Figure 5.17 Fixed-end reaction forces and moments.



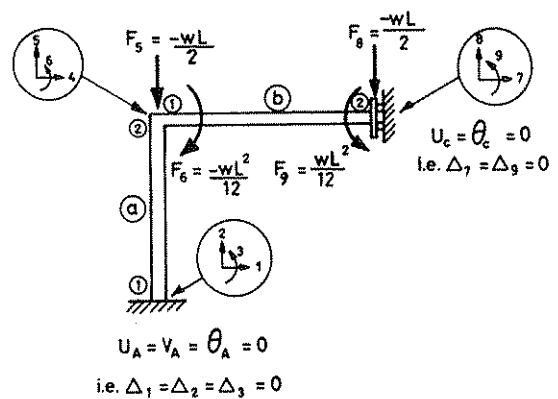
(a) STRUCTURE

$$A = 1000 \text{ mm}^2 = 10^{-3} \text{ m}^2$$

$$E = 200 \text{ GPa}$$

$$L = 2 \text{ m}$$

$$w = 12 \text{ kN/m}$$



(b) STRUCTURAL MODEL

Figure 5.18 Sample problem with distributed load.

The solution will be presented in terms of the seven steps of Table 5.1.

Step 1. Define the Structural Model

The structural modeling is illustrated in Fig. 5.18b. Because of symmetry only half of the structure is modeled. To permit this a node is introduced at point C, even though there is no joint there. The geometric conditions arising from symmetry are: zero horizontal displacement and zero slope at the point of symmetry. The clamped support at A means that the three displacements at that point are also zero. Although the frame is orthogonal the solution will be performed using the ordinary beam element, so that the axial degree of freedom is included in the modeling. The result will then be compared with a solution obtained using the flexure-only beam element. The structure degrees of freedom are numbered from 1 to 9 as shown. The uniform distributed load of intensity w is represented by means of the statically equivalent loads $wL/2$ and $wL^2/12$. It may be seen that F_9 will not actually be applied to the structure because degree of freedom 9 is eliminated due to symmetry. The axial and flexural stiffness parameters are $\alpha = AE/L$ and $b = EI/L^3$. Their ratio is

$$\frac{\alpha}{b} = \frac{EA}{L} \frac{L^3}{EA\rho^2} = \left(\frac{L}{\rho}\right)^2$$

where ρ = radius of gyration of the beam cross section. The ratio L/ρ is the *slenderness ratio* of the beam and for typical flange-and-web beams its value is between 10 and 40. For illustrative purposes we take a deliberately low value of 10, which then means that $\alpha/b = 100$. The numerical values of all dimensions and physical properties are given in the figure.

Step 2. Calculate Each Element Stiffness Matrix in Structure Coordinates

The general expression for the beam element stiffness matrix at an arbitrary orientation χ is given in (5.3.16). The values of λ and μ for the two elements are as follows.

Member	χ	λ	μ
Ⓐ	90°	0	1
Ⓑ	0°	1	0

The element stiffness matrix for element Ⓐ is given. It is suggested that the reader should evaluate at least a few of the terms himself and then compare them with the following.

$$K^{\text{Ⓐ}} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{bmatrix} 12b & 0 & -6bL & -12b & 0 & -6bL \\ 0 & \alpha & 0 & 0 & -\alpha & 0 \\ -6bL & 0 & 4bL^2 & 6bL & 0 & 2bL^2 \\ -12b & 0 & 6bL & 12b & 0 & 6bL \\ 0 & -\alpha & 0 & 0 & \alpha & 0 \\ -6bL & 0 & 2bL^2 & 6bL & 0 & 4bL^2 \end{bmatrix} \end{matrix}$$

$$= \begin{matrix} & \begin{matrix} \times & \times & \times & 4 & 5 & 6 \end{matrix} \\ \begin{matrix} \times \\ \times \\ \times \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{bmatrix} & & & & & \\ & & & & & \\ & & & & & \\ & & & 12 & 0 & 12 \\ & & & 0 & 100 & 0 \\ & & & 12 & 0 & 16 \end{bmatrix} \end{matrix}$$

The location vector is written above the matrix. In a hand solution such as this it is often worthwhile to anticipate which degrees of freedom are to be eliminated and to not bother calculating these terms in K^e . For greater clarity the foregoing matrix has been written out in full initially but in substituting the numerical values this short-cut has been used. The eliminated degrees of freedom are indicated by $\times$. The stiffness matrix for element Ⓑ is

$$K^{\text{Ⓑ}} = \begin{matrix} & \begin{matrix} 4 & 5 & 6 & 7 & 8 & 9 \end{matrix} \\ \begin{matrix} 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{matrix} & \begin{bmatrix} \alpha & 0 & 0 & -\alpha & 0 & 0 \\ 0 & 12b & 6bL & 0 & -12b & 6bL \\ 0 & 6bL & 4bL^2 & 0 & -6bL & 2bL^2 \\ -\alpha & 0 & 0 & \alpha & 0 & 0 \\ 0 & -12b & -6bL & 0 & 12b & -6bL \\ 0 & 6bL & 2bL^2 & 0 & -6bL & 4bL^2 \end{bmatrix} \end{matrix}$$

$$= \begin{matrix} & \begin{matrix} 4 & 5 & 6 & \times & 8 & \times \end{matrix} \\ \begin{matrix} 4 \\ 5 \\ 6 \\ \times \\ 8 \\ \times \end{matrix} & \begin{bmatrix} 100 & 0 & 0 & -100 & 0 & 0 \\ 0 & 12 & 12 & 0 & -12 & 12 \\ 0 & 12 & 16 & 0 & -12 & 8 \\ -100 & 0 & 0 & 100 & 0 & 0 \\ 0 & -12 & -12 & 0 & 12 & -12 \\ 0 & 12 & 8 & 0 & -12 & 16 \end{bmatrix} \end{matrix}$$

Steps 3 and 4. Assemble the Structure Stiffness Matrix and Apply the Boundary Conditions

In this problem these two steps are combined because the elimination of the degrees of freedom is here being achieved by simply not assembling these terms in the structure stiffness matrix. If a given structure is to be analyzed repeatedly with the same boundary condi-

tions (as, for example, in structural optimization) then this technique is more efficient.

The result of the assembly process is given next. Here again it is suggested that the reader should perform the assembly process himself and then compare his result with that given.

$$\mathbf{K} = \begin{bmatrix} & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{matrix}$$

$$\begin{bmatrix} & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{matrix}$$

Thus the equilibrium equations are a 4×4 system of equations. The load vector must be expressed in units which are consistent with $\mathbf{K}$. In this problem the units of α and b , and hence of $\mathbf{K}$, are MN/m. Therefore the magnitudes of the equivalent nodal loads are

$$\frac{wL}{2} = \frac{12 \times 10^{-3}(2)}{2} = 12 \times 10^{-3} \text{ MN}$$

and

$$\frac{wL^2}{12} = \frac{12 \times 10^{-3}(4)}{12} = 4 \times 10^{-3} \text{ MNm}$$

Hence the system of equations is

$$\begin{bmatrix} 0 \\ -12 \\ -4 \\ -12 \end{bmatrix} \times 10^{-3} = \begin{bmatrix} 112 & 0 & 12 & 0 \\ 0 & 112 & 12 & -12 \\ 12 & 12 & 32 & -12 \\ 0 & -12 & -12 & 12 \end{bmatrix} \begin{bmatrix} \Delta_4 \\ \Delta_5 \\ \Delta_6 \\ \Delta_8 \end{bmatrix}$$

Step 5. Solve for Displacements and Reactions

The solution to this system is

$$\Delta_4 = 0.0917 \times 10^{-3} \text{ m}$$

$$\Delta_5 = -0.2400 \times 10^{-3} \text{ m}$$

$$\Delta_6 = -0.8550 \times 10^{-3} \text{ rad}$$

$$\Delta_8 = -2.0950 \times 10^{-3} \text{ m}$$

Reaction Forces. If desired, the support reactions, could now be calculated from the appropriate (excluded) equilibrium equations. For example F_3 , the

reaction moment at A, is given by the third excluded equation

$$F_3 = b[-6L \ 0 \ 4L^2 \ 6L \ 0 \ 2L^2 \ 0 \ 0 \ 0] \Delta$$

$$= -5.74 \times 10^{-3} \text{ MNm}$$

Steps 6 and 7. Calculate the Element Internal Forces

A. Element (a)

1. Element Nodal Displacements

For a simple structure such as this the element nodal displacements may be obtained from the structure displacements simply by inspection, but for a more complex structure, or in a computer program, it would be necessary to use the location vector for the allocation of the Δ values and then to use the transformation matrix $\mathbf{T}$ to obtain the element nodal displacements in element coordinates. For example, for element (a)

$$\delta_a = \mathbf{T}_a \Delta_a = \begin{bmatrix} \lambda & \mu & 0 & 0 & 0 & 0 \\ -\mu & \lambda & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda & \mu & 0 \\ 0 & 0 & 0 & -\mu & \lambda & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \Delta_1 \\ \Delta_2 \\ \Delta_3 \\ \Delta_4 \\ \Delta_5 \\ \Delta_6 \end{bmatrix}$$

and since $\lambda = 0$ and $\mu = 1$ the result is

$$\delta_a = \begin{bmatrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \end{bmatrix}_a = \begin{bmatrix} \Delta_2 \\ -\Delta_1 \\ \Delta_3 \\ \Delta_5 \\ -\Delta_4 \\ \Delta_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -0.24 \\ -0.0916 \\ -0.8550 \end{bmatrix} \times 10^{-3}$$

2. Element Internal Forces

For the beam element the internal forces are axial stress, shear force, and bending moment. Since the first two are constant and the third is linear within the element it is sufficient to calculate these forces only at the nodes. The expression for this, in terms of nodal displacements, is given by (5.3.18). The results are

Axial stress

$$\sigma_{x1} = \sigma_{x2} = \frac{E}{L}(-u_1 + u_2)$$

$$= \frac{200 \times 10^3}{2}(0 - .24 \times 10^{-3}) = -24 \text{ MPa}$$

Shear force

$$Q_1 = Q_2 = b[0 \quad 12 \quad 6L \quad 0 \quad -12 \quad 6L] \delta_{\text{e}}$$

$$= -9.16 \times 10^{-3} \text{ MN} = -9.16 \text{ kN}$$

Bending moment

$$\begin{Bmatrix} M_z(0) \\ M_z(L) \end{Bmatrix} = b \begin{bmatrix} 0 & -6L & -4L^2 & 0 & 6L & -2L^2 \\ 0 & 6L & 2L^2 & 0 & -6L & 4L^2 \end{bmatrix} \delta_{\text{e}}$$

which gives

$$M_z(0) = 5.74 \text{ kNm}$$

$$M_z(L) = -12.58 \text{ kNm}$$

The distribution of $M_z(x)$ is plotted in Fig. 5.19a to the left of the element.

3. Element Nodal Forces

Usually there is no need to calculate the element nodal forces f but we will do so in this example to illustrate the distinction between nodal forces F_{y1} , F_{y2} and shear force Q , and between nodal moments M_{z1} , M_{z2} and

bending moment $M_z(x)$. It will also be demonstrated that each element is in equilibrium. The quantity that relates element nodal forces and element nodal displacements is the element stiffness matrix, since $f = k^e \delta$. Hence from (5.3.15)

$$F_{x1} = \alpha(u_1 - u_2) = 100(0 + 0.24 \times 10^{-3})$$

$$= 24 \text{ kN}$$

$$F_{y1} = b(12v_1 + 6L\theta_1 - 12v_2 + 6L\theta_2)$$

$$= -9.16 \text{ kN}$$

$$M_{z1} = b(6Lv_1 + 4L^2\theta_1 - 6Lv_2 + 2L^2\theta_2)$$

$$= -5.74 \text{ kNm}$$

$$F_{x2} = \alpha(-u_1 + u_2) = -24 \text{ kN}$$

$$F_{y2} = b(-12v_1 - 6L\theta_1 + 12v_2 - 6L\theta_2)$$

$$= +9.16 \text{ kN}$$

$$M_{z2} = b(6Lv_1 + 2L^2\theta_1 - 6Lv_2 + 4L^2\theta_2)$$

$$= -12.58 \text{ kNm}$$

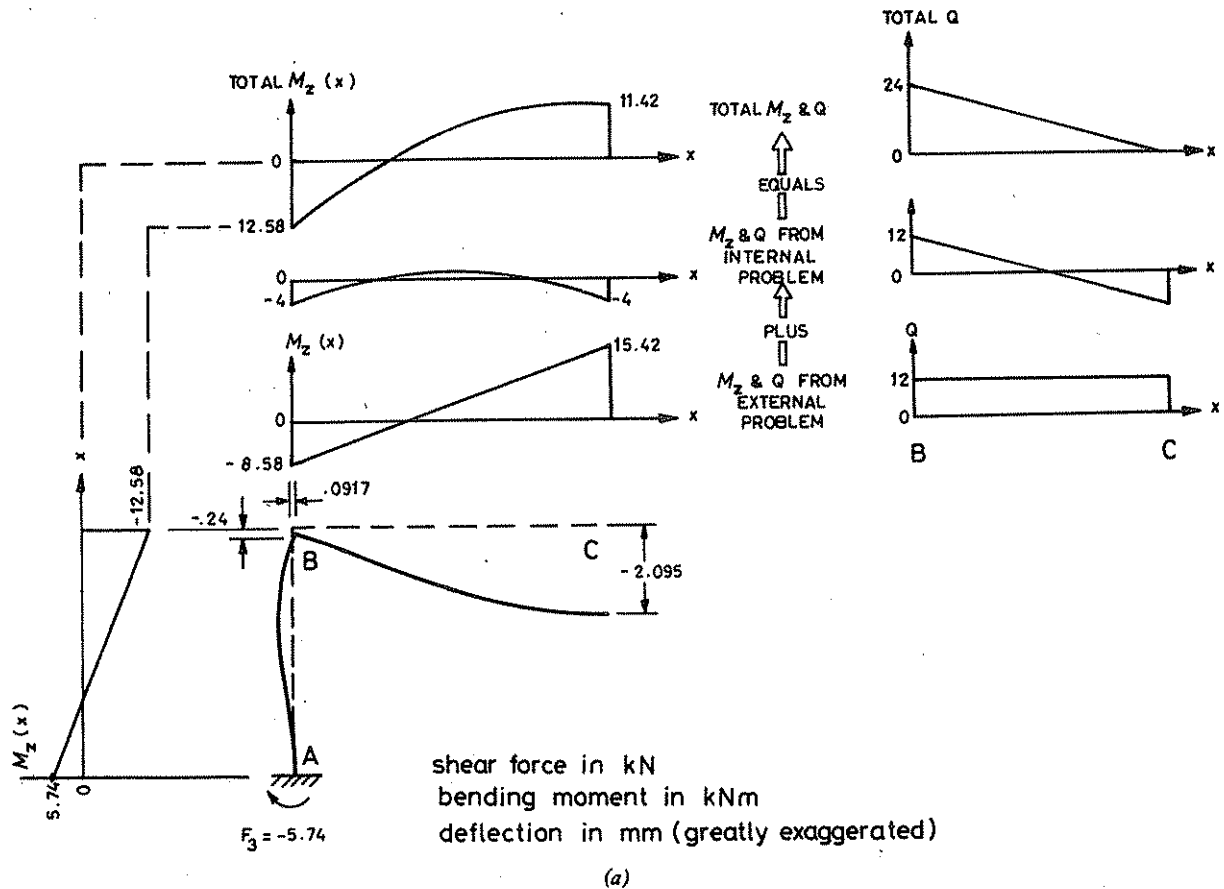


Figure 5.19a Deflection, shear force, and bending moment.

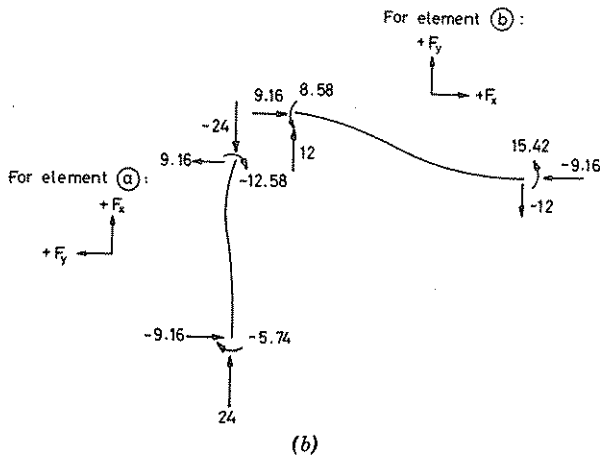


Figure 5.19b Element nodal forces.

These forces and moments are illustrated in Fig. 5.19b. Equilibrium of forces is obvious, and equilibrium of moments may be verified by taking moments about point A:

$$\sum M_A = -5.74 - 12.58 + 9.16L = 0$$

B. Element (b)

1. Element Nodal Displacements

In a similar fashion the nodal displacements of element (b) are

$$\delta_{(b)} = \begin{Bmatrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \end{Bmatrix}_{(b)} = \begin{Bmatrix} \Delta_4 \\ \Delta_5 \\ \Delta_6 \\ \Delta_7 \\ \Delta_8 \\ \Delta_9 \end{Bmatrix} = \begin{Bmatrix} 0.0916 \\ -0.2400 \\ -0.8550 \\ 0 \\ -2.0950 \\ 0 \end{Bmatrix} \times 10^{-3}$$

2. Element Internal Forces

From (5.3.18)

Axial Stress

$$\begin{aligned} \sigma_{x1} = \sigma_{x2} &= \frac{E}{L}(-u_1 + u_2) \\ &= \frac{200 \times 10^3}{2}(-0.0916 \times 10^{-3} + 0) \\ &= -9.16 \text{ MPa} \end{aligned}$$

Shear force

$$\begin{aligned} Q_1 = Q_2 &= b[0 \quad 12 \quad 6L \quad 0 \quad -12 \quad 6L] \delta_{(b)} \\ &= 12 \times 10^{-3} \text{ MN} = 12 \text{ kN} \end{aligned}$$

Bending moment

$$\begin{aligned} \begin{Bmatrix} M_x(0) \\ M_x(L) \end{Bmatrix} &= b \begin{bmatrix} 0 & -6L & -4L^2 & 0 & 6L & -2L^2 \\ 0 & 6L & 2L^2 & 0 & -6L & 4L^2 \end{bmatrix} \delta_{(b)} \\ &= \begin{Bmatrix} -8.58 \\ 15.42 \end{Bmatrix} \text{ kNm} \end{aligned}$$

The distributions of Q and of $M_x(x)$ are plotted in Fig. 5.19a, immediately above the element. Since this element has a distributed load, these two internal forces must be corrected by superimposing the corresponding forces from the solution of the internal problem. In the present example the internal problem is simply a uniformly loaded, clamped beam. The corresponding shear force and bending moment distributions are shown just above those obtained from the frame analysis. The final and complete distributions, which are the sum of these two, are drawn at the top of the figure. Note that the total value of M_x at point B, -12.58 , agrees with the value which was calculated in element (a). This is another useful check on the solution.

3. Element Nodal Forces

Here again, although not necessary for the solution, we shall calculate the element nodal forces in order to point out some of the relationships and differences between these and other quantities. As before, $f_{(b)} = k^e \delta_{(b)}$ with k^e given by (5.3.15). Substitution of $\delta_{(b)}$ gives

$$F_{x1} = -F_{x2} = 9.16$$

$$F_{y1} = -F_{y2} = 12$$

$$M_{z1} = 8.58; \quad M_{z2} = 15.42$$

These forces and moments are illustrated in Fig. 5.19b. Once again, equilibrium of forces is immediately evident and equilibrium of moments can be verified by taking moments about point B:

$$\sum M_B = 8.58 + 15.42 - 12L = 0.$$

This summation of moments illustrates the fact that frame analysis takes no account of any eccentricity of axial loads arising from the beam's deflection. This procedure is valid as long as the deflections are small, which they are in this example because the largest deflection is 2.095 mm, which is only 0.1% of the beam length L . The small-deflection assumption is usually valid for the elastic analysis of ships and other plated steel frame structures, because the overall stiffness of such structures is usually sufficiently large

such that any load which does not cause yielding produces only small deflections. However, the stiffness of a structure depends on the material stiffness, on the member proportions and scantlings (thickness, etc.) and on the arrangement of the members. Hence when using frame analysis (or any other elastic, small-deflection technique) a check should always be made on the size of the deflections relative to the dimensions of the structure.

Another observation which may be made in the foregoing example is that at point B the net vertical force of -12 kN is equal to the applied load at that point, which is the equivalent load, $F_4 = -wL/2$. Similarly, the net moment of -4 kNm is equal to the applied moment, $F_6 = -wL^2/12$. However, the net moment at point C (15.42 kNm) is not equal to the equivalent moment, $F_9 = wL^2/12$, because the latter was never applied to the structure; it was excluded by the zero-slope boundary condition at point C.

SOLUTION USING FLEXURE-ONLY BEAM ELEMENT

In the foregoing example the ratio of α and b is 100. In more slender frames and particularly in plated frameworks such as occur in ships the ratio is much larger still, often being 500 or more. If such frames are completely orthogonal, such that axial loads and displacements are independent (or nearly independent) of lateral loads and displacements, and if axial forces and displacements are not of primary interest, then the axial degree of freedom can be ignored and the flexure-only beam element may be used. The use of this element is equivalent to regarding the ratio α/b as being infinity, such that there is no shortening or elongation of any beam. If several of these elements are coaxial and if any one of the nodes is externally restrained against axial displacement, this restraint will extend to all nodes on that axis. Moreover, if another beam which is similarly restrained is attached to one of these nodes and is orthogonal to it, then the orthogonal deflection of that node is also restrained. Thus in an orthogonal framework there may be many nodes in which one or more translational degrees of freedom are restrained; the number of such restraints depends on the arrangement of the elements and on the extent of external restraint which is imposed on the structure. All of these restrained degrees of freedom can be eliminated, and this can give a significant reduction in the size of the overall problem. In an *ad hoc* solution formulated for a specific structure (as opposed to using a general-purpose frame analysis program) these re-

strained degrees of freedom can be omitted from the structural model right from the beginning, and this is particularly helpful in such solutions, especially if a quick and simple estimate is being sought.

For example, in the above portal frame the vertical restraint at A, in conjunction with nonshortening of element (a), causes point B to also have vertical restraint. Likewise, the horizontal restraint at C is transmitted to point B by element (b). Hence at point B there is only one degree of freedom: rotation. The structural model is therefore reduced to the extremely simple, two-degree-of-freedom problem shown in Fig. 5.20. The solution is

$$\Delta_1 = -0.8 \times 10^{-3} \text{ rad}$$

$$\Delta_2 = -1.8 \times 10^{-3} \text{ m}$$

These agree reasonably well with the displacements from the first solution (-0.855×10^{-3} and -2.095×10^{-3}). The difference in the values of Δ_2 is almost entirely accounted for by the axial shortening of element (a) (-0.24×10^{-3}). In frame structures the response that is usually of greatest interest is bending moment. For the flexure-only element this is given by (5.3.12) and for the given structure the results are:

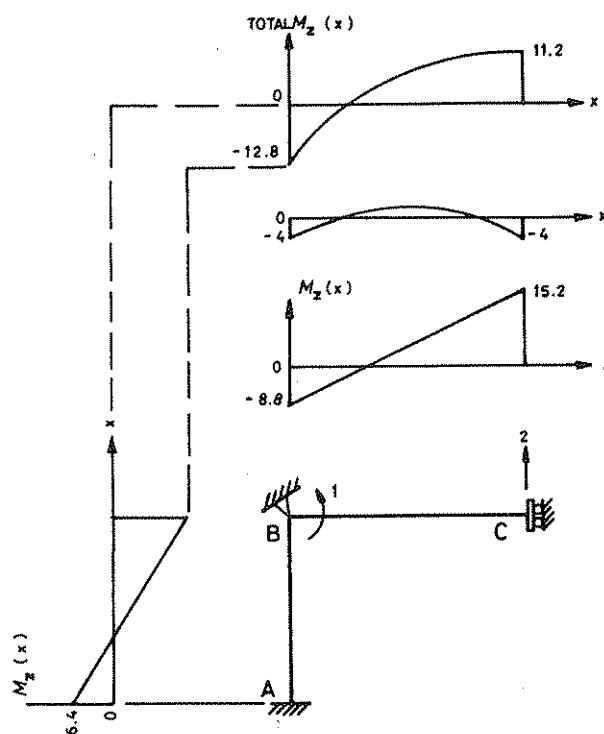


Figure 5.20 Solution using flexure-only beam element.

Element (a)

$$\begin{aligned} \begin{Bmatrix} M_z(0) \\ M_z(L) \end{Bmatrix} &= S\delta_a \\ &= b \begin{bmatrix} -6L & -4L^2 & 6L & -2L^2 \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix} \begin{Bmatrix} 0 \\ 0 \\ 0 \\ -0.8 \end{Bmatrix} \\ &= \begin{Bmatrix} 6.4 \\ -12.8 \end{Bmatrix} \end{aligned}$$

Element (b)

$$\begin{aligned} \begin{Bmatrix} M_z(0) \\ M_z(L) \end{Bmatrix} &= S\delta_b = S \begin{Bmatrix} 0 \\ -0.8 \\ -1.8 \\ 0 \end{Bmatrix} \\ &= \begin{Bmatrix} -8.80 \\ 15.20 \end{Bmatrix} \end{aligned}$$

This result for element (b) is not complete because this element carries a distributed load and the above bending moment corresponds only to the external problem. The internal problem is the same as before—a uniformly loaded clamped beam. The final solution is the sum of the two and this is shown in Fig. 5.20. The values of bending moment agree very well with those of the earlier, more exact solution, particularly in those locations where the bending is large, as can be seen from the following comparison:

Location	Ordinary Beam Element	Flexure-Only Beam Element	% Difference
A	5.74	6.4	11.5
B	12.58	12.8	1.8
C	11.42	11.2	1.9

In this example a relatively low value of 100 was chosen for the ratio α/b . With larger values the accuracy of the flexure-only beam element would be still greater.

5.4 APPLICATION TO GRILLAGES AND Laterally LOADED PANELS

A grillage is a plane structure of intersecting beams, with or without plating, which carries a lateral load through the action of beam bending. Most grillages are orthogonal; that is, they are made up of two sets of

parallel beams which intersect at right angles. Throughout this text the term "grillage" implies a flat grillage. It also implies that in-plane loads are sufficiently small such that all in-plane effects (both load and response) are completely independent of the lateral loads and the grillage bending response. If this condition is not satisfied then the structure will be referred to as a cross-stiffened panel. The response under in-plane loads is dealt with in Chapters 13 and 14. Some examples of grillages are platform decks and decks near the ship's neutral axis, transverse bulkheads, ramps, and hatch covers.

A grillage may be either regular or irregular. A regular grillage is rectangular in shape, the beams in each direction are of uniform size and spacing and each beam is prismatic and continuous. For regular grillages several special analysis methods are available which take advantage of the regularity; see, for example, Refs. 1 and 2.

If a grillage cannot be idealized as two orthogonal sets of identical full-length prismatic beams intersecting each other, then it is irregular and it must be modeled as a laterally loaded plane frame made up of many individual flexure-only beam elements which are connected at nodes. Each of these elements may be different and they need not be orthogonal. Thus, because of the great versatility of frame analysis even quite complex grillages are no more than straightforward applications of this method. For this reason, if an efficient frame analysis computer program is readily available it is often easier and more efficient (because of familiarity with the program) to use it for all grillages—regular as well as irregular.

Allocation of Distributed Load

In stiffened panels the plating transmits the lateral load to the beams and in cross-stiffened panels it is necessary to estimate the proportion of the load which is taken by each set of beams. When one set of beams (say the stiffeners) is more closely spaced than the other (the girders) it is reasonable to assume that all of the load is transmitted to the stiffeners. The girders then support the stiffeners, and so the loads on the girders are the reaction forces at each stiffener. For a uniform pressure p and a stiffener spacing b , the distributed load q on each stiffener is $q = pb$.

When the spacings are more nearly equal it is necessary to apportion the load between the two sets of beams. An empirical but sufficiently accurate allocation is given by Faulkner's equation [3]:

$$\left. \begin{aligned} \text{load on each short} &= pb^2 \\ \text{adjacent beam} \\ \text{load on each} &= pb^2 \left(\alpha - \frac{1}{2} \right) \\ \text{long adjacent beam} \end{aligned} \right\} \quad (5.4.1)$$

in which $\alpha = a/b$

a = longer dimension of the plating
between the beams

b = shorter dimension of the plating
between the beams

Strictly speaking, the distribution of these loads along each of the beams would give a triangular and a trapezoidal load curve, respectively, but these are usually simplified to uniform distributions q_a and q_b :

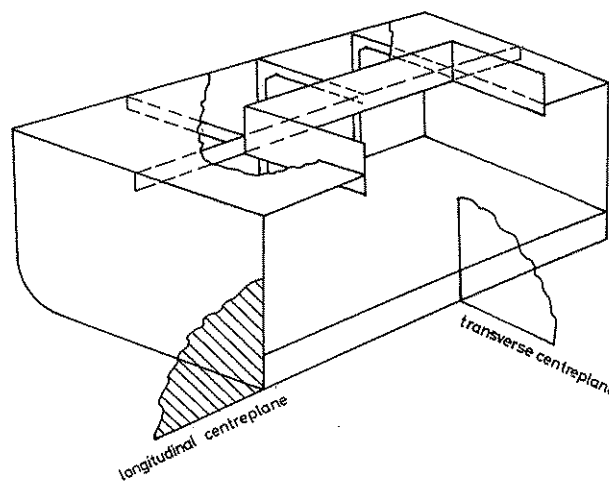
$$\left. \begin{aligned} q_a &= pb \left(1 - \frac{1}{2\alpha} \right) \\ q_b &= \frac{pb}{2} \end{aligned} \right\} \quad (5.4.2)$$

Structural Modeling of a Grillage

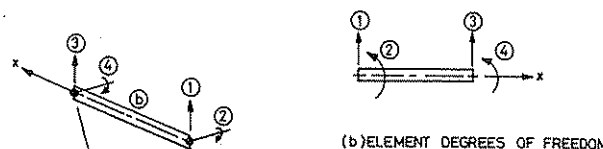
The analysis of grillages is simplified by the fact that the load, and hence the displacements and rotations, all occur in a plane normal to the structure. By definition all in-plane effects are absent or are small enough to be calculated separately and superimposed. Therefore the flexure-only beam element is ideal for grillage analysis. If the grillage is orthogonal, and most are, then there are only three degrees of freedom at each node: vertical deflection and two rotations. Also, if the orientation of the element in the structure is properly chosen then the transformation from element to structure coordinates is not required; that is, each element stiffness matrix k^e can be inserted directly into K , as indicated by the location vector. This is made possible by choosing the orientation of each element (i.e., the direction of the element x -axis) such that the four degrees of freedom of the element act in the same directions as the corresponding degrees of freedom of the structure.

The deck grillage of Fig. 5.21a will be used as an example to illustrate the principal steps in this type of application of frame analysis. We will also take advantage of this example to illustrate several useful modeling techniques and approximations:

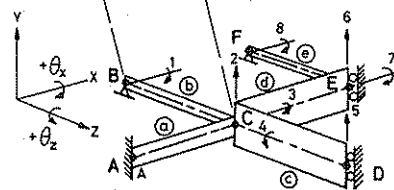
1. As noted in Section 5.3, the nodes of beam elements are located at the neutral axis of the beam. In the



(a) STRUCTURE



(b) ELEMENT DEGREES OF FREEDOM



(c) STRUCTURAL MODEL

Figure 5.21 Grillage example.

structural model of Fig. 5.21c the nonalignment of the element is ignored and they are joined directly together, as if the neutral axes of all elements were in the same plane. This vertical repositioning of members is valid only if there are no (appreciable) axial loads, because otherwise the eccentricity of the connections would give rise to further moments which have not been allowed for. Therefore let us assume that the deck in Fig. 5.21 is a "platform deck" (not longitudinally continuous).

2. Torsional stiffness of the beams is ignored because this type of stiffness is relatively small for thin-wall beams of open cross section. This simplifies the analysis because it reduces the number of degrees of freedom in the structure. Also, the slight error which is incurred lies on the conservative side.

3. Support of the deck beams at the side shell is treated as simple support because the side frames provide relatively little rotational restraint. Again, the

error in this approximation lies on the conservative side. If desired the side frames could be included in the grillage because their vertical orientation does not affect their flexural properties.

4. Support of the longitudinal hatch side girder at the transverse bulkhead is treated as clamped because the hatch side girder is longitudinally continuous through the bulkhead. This is done only for simplicity; if desired the grillage could extend over several cargo holds.

A typical flexure-only beam element is shown in Fig. 5.21b. In this example the element degrees of freedom are circled in order to distinguish them from the structure degrees of freedom. The structural model for the grillage is shown in Fig. 5.21c, showing the orientation of the various elements in the structure. The orientation of each element is chosen such that the element rotational degrees of freedom have the same positive rotational direction as the rotational degrees of freedom of the structure. Thus, for example, element (b) must be inserted into the structure such that its x -axis points in the $-Z$ -direction in order that its rotational degrees of freedom (numbers ② and ④) may act in the same direction as the corresponding structure degrees of freedom (numbers 3 and 1) for which the positive direction is θ_x , as shown in the figure.

The elimination of structural degrees of freedom due to boundary conditions (structural supports and symmetry) is done as part of the setting up of the structural model. The simple support at the side shell eliminates vertical deflection and twist at B and F; the clamped support at the bulkhead eliminates all deflections and rotations at A; transverse symmetry requires zero slope at D and hence the bending rotation is eliminated there, and longitudinal symmetry eliminates bending rotation at E. In addition, since the torsional stiffness is not being modeled, it is permissible to eliminate the torsional degree of freedom at points D and E. This may at first seem to be an artificial strengthening of the structure but this is not so

because the absence of any torsional stiffness in the beam elements means that there is no way for this restraint to be transmitted into the structure; it has no effect whatever. In fact, the absence of torsional stiffness in the element makes it mandatory to eliminate the torsional degree of freedom at points D and E, and also at points A, B, and F, because none of the beams can offer any resistance to a torsional displacement and so the structure stiffness matrix would be singular, having zeros on the main diagonal at these degree of freedom positions.

After these eliminations have been made there are a total of eight structural degrees of freedom and these are numbered as shown in Fig. 5.21c. The numbering is chosen with a view to keeping the bandwidth of the equilibrium equations as narrow as possible, but the chosen numbering sequence is not unique; there are several others which would have been equally satisfactory.

The need to place the rotational degrees of freedom of the elements in alignment with those of the structure means that for elements (a) and (d) the x -axis points in the $+X$ -direction, and for elements (b), (c) and (e) the x -axis points in the $-Z$ -direction. Therefore the location vectors for the five elements are

	①	②	③	④
(a)	0	0	2	4
(b)	2	3	0	1
(c)	5	0	2	3
(d)	2	4	6	0
(e)	6	7	0	8

These vectors indicate precisely where each term of each element stiffness matrix k^e is to be inserted into K . For brevity we adopt the symbol a_{ij} to represent the term $k_{ij}^{(a)}$ in the element stiffness matrix for element (a), and a similar notation for elements (b), (c), and (d). The resulting structure stiffness matrix is

	1	2	3	4	5	6	7	8
1	b_{44}	b_{41}	b_{42}					
2	b_{14}	$a_{33} + b_{11}$ $+ c_{33} + d_{11}$	$c_{34} + b_{12}$	$a_{34} + d_{12}$	c_{31}	d_{13}		
3	b_{24}	$c_{43} + b_{21}$	$c_{44} + b_{22}$		c_{41}			
4		$a_{43} + d_{21}$		$a_{44} + d_{22}$		d_{23}		
5		c_{13}	c_{14}		c_{11}			
6		d_{31}		d_{32}		$d_{33} + e_{11}$	e_{12}	e_{14}
7						e_{21}	e_{22}	e_{24}
8						e_{41}	e_{42}	e_{44}

The next step would be to insert the loads and then solve the equilibrium equations. Thus by means of frame analysis the analysis of a relatively complex grillage structure has been reduced to a symmetric, banded 8×8 system of equations, which is small enough to be solved on a hand-held programmable calculator.

5.5 GENERAL BEAM ELEMENT

Thus far, for simplicity, we have considered beam bending in only one plane and we have ignored torsion and the effect of shear deflection. In this section these additional features of beam bending are discussed and are combined to form a stiffness matrix for a general beam element which is suitable for most three-dimensional frames, especially the plated steel frameworks found in ships.

The beam is assumed to be a straight beam of uniform cross section capable of resisting axial forces, bending moments about the two principal axes in the plane of its cross section, and twisting moments about its centroidal axis. The nodal forces acting on the beam are: axial forces f_1 and f_7 ; shearing forces f_2, f_3, f_8 , and f_9 ; moments f_5, f_6, f_{11} , and f_{12} ; and twisting moments (torques) f_4 and f_{10} . The location and positive direction of these forces are shown in Fig. 5.22. The corresponding displacements $\delta_1, \dots, \delta_{12}$ will be taken, as before, to be positive in the positive directions of the forces.

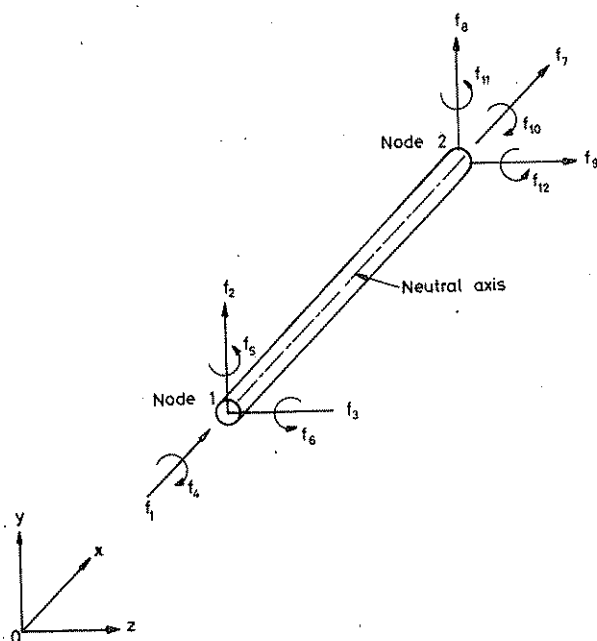


Figure 5.22 General beam element.

As is well known, the axial forces f_1 and f_7 depend only on their corresponding displacements; the same is true of the torques f_4 and f_{10} . However, for arbitrary orientation of the y - and z -axes the bending moments and shearing forces in the xy -plane would depend not only on their corresponding displacements but also on the displacements corresponding to the forces in the xz -plane. Only if the x -axis coincides with the centroidal axis of the beam and the xy - and xz -planes coincide with the principal axes of the cross section can the bending and shear in the two planes be considered independently of each other. Hence we adopt this orientation for the element axes.

Another advantage of using principal axes is that this allows the element stiffness matrix (which is of order 12×12) to be expressed in a more compact and banded form, in terms of four small submatrices each of which refers exclusively to a particular mode of displacement: axial displacement, u ; y displacement, v ; z displacement, w ; and torsional displacement, θ . This bandedness bears no relationship to the speed of solution because the element stiffness matrix is merely a record of the element's properties, and the order of terms within it is arbitrary. However, the compactness makes this form more convenient for derivations and for observing relationships between the various modes of displacement. The compact form is obtained by using a different numbering system for the element degrees of freedom. Instead of numbering all of the degrees of freedom at one node before proceeding to the other node they are divided into the above four groups (u, v, w , and θ displacements) and each group is numbered fully before proceeding to the next group. These alternative degrees of freedom are related to the usual degrees of freedom δ of Fig. 5.22 as follows:

$$\left. \begin{aligned} \mathbf{u} &= \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} \delta_1 \\ \delta_7 \end{Bmatrix} \\ \mathbf{v} &= \begin{Bmatrix} v_1 \\ v'_1 \\ v_2 \\ v'_2 \end{Bmatrix} = \begin{Bmatrix} \delta_2 \\ \delta_6 \\ \delta_8 \\ \delta_{12} \end{Bmatrix} & \mathbf{w} &= \begin{Bmatrix} w_1 \\ w'_1 \\ w_2 \\ w'_2 \end{Bmatrix} = \begin{Bmatrix} \delta_3 \\ \delta_5 \\ \delta_9 \\ \delta_{11} \end{Bmatrix} \\ \theta &= \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = \begin{Bmatrix} \delta_4 \\ \delta_{10} \end{Bmatrix} \end{aligned} \right\} \quad (5.5.1)$$

Shear deflection occurs in the v and w directions, and since these two are quite similar we shall concentrate on the former and derive the generalized element stiffness (sub)matrix $\mathbf{k}_v$ for the v and v' degrees of freedom (i.e., for deflection in the y -direction and bending about the z -direction) taking into account the effect of shear deflection in the y -direction.

Effect of Shear Deflection

If a beam has a large value of bending stiffness parameter b ($b = EI/L^3$) because of large flanges or because it is short or because it has a deep web (say one-fifth of the length or more) then the bending deflection is small and the vertical shear deflection of the web (which is ignored in elementary beam theory) can be of the same order as the bending deflection, such that it cannot be ignored. We therefore seek to derive an expression for the beam's stiffness in regard to vertical (i.e., y -directed) displacement which will account for this effect. To do so we will use the same technique as for the bar element in Section 5.1; that is, impose each of the nodal displacements in turn while all of the others are held at zero and determine all of the nodal forces which are set up. Each of the nodal forces is then divided by the single nodal displacement and the quotients thus obtained are the terms in one complete column of the element stiffness matrix—the column corresponding to that displacement.

VERTICAL DISPLACEMENT AND ROTATION (v AND v')

The vertical deflection v of a beam subjected to end (nodal) forces and moments, as shown in Fig. 5.23a, is given by

$$v = v_b + v_s$$

where v_b is the vertical deflection due to bending strains and v_s is the additional deflection due to shearing strains, such that

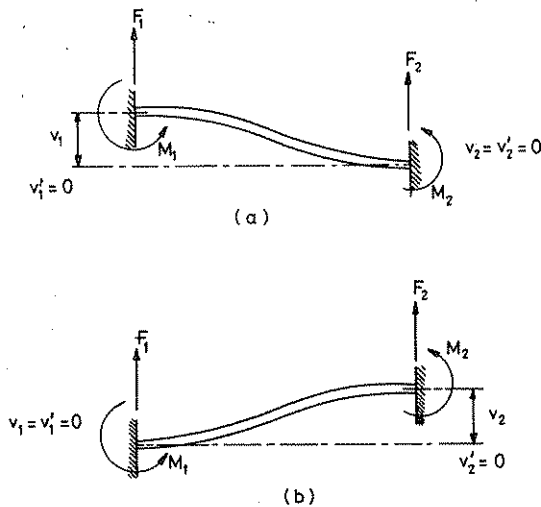


Figure 5.23 Nodal displacements imposed individually.

$$\frac{dv_s}{dx} = \frac{-F_1}{GA_s} \quad (5.5.2)$$

with A_s representing the beam cross-sectional area effective in shear. The bending deflection for the beam shown in Fig. 5.23a is governed by the differential equation

$$EI_z \frac{d^2 v_b}{dx^2} = F_1 x - M_1 \quad (5.5.3)$$

From integration of (5.5.2) and (5.5.3) it follows that

$$EI_z v = \frac{F_1 x^3}{6} - \frac{M_1 x^2}{2} + \left(C_1 - \frac{F_1 EI_z}{GA_s} \right) x + C_2 \quad (5.5.4)$$

where C_1 and C_2 are the constants of integration. The component of slope due to shear, dv_s/dx , is not a rotation in the ordinary sense and is totally unrelated to the nodal moments. Therefore the second or rotational degree of freedom is the slope due to bending only: $v' = dv_b/dx$. Hence the boundary condition for a clamped end is $dv_b/dx = 0$, and the total slope at this point is nonzero and is equal to dv_s/dx .

Therefore the boundary conditions in Fig. 5.23a are:

$$\frac{dv}{dx} = \frac{dv_s}{dx} = \frac{-F_1}{GA_s} \quad \text{at } x = 0, x = L$$

$$\text{and} \quad v = 0 \quad \text{at } x = L$$

Equation 5.5.4 becomes

$$EI_z v = \frac{F_1 x^3}{6} - \frac{M_1 x^2}{2} - \frac{F_1 \Phi x L^2}{12} + (1 + \Phi) \frac{L^3 F_1}{12} \quad (5.5.5)$$

$$\text{where} \quad \Phi = \frac{12EI_z}{GA_s L^2}$$

and the forces are related by

$$M_1 = \frac{F_1 L}{2}$$

The remaining forces acting on the beam can be determined from the equations of equilibrium:

$$\left. \begin{aligned} F_2 &= -F_1 \\ M_2 &= -M_1 + F_1 L \end{aligned} \right\} \quad (5.5.6)$$

and

Now at $x = 0$, $v = v_1$ and hence from (5.5.5)

$$v_1 = (1 + \Phi) \frac{F_1 L^3}{12EI_z}$$

Using the foregoing equations the following results are obtained

$$k_{11} = \frac{F_1}{v_1} = \frac{12EI_z}{(1 + \Phi)L^3}$$

$$k_{21} = \frac{M_1}{v_1} = \frac{F_1 L}{2v_1} = \frac{6EI_z}{(1 + \Phi)L^2}$$

$$k_{31} = \frac{F_2}{v_1} = \frac{-12EI_z}{(1 + \Phi)L^3}$$

$$k_{41} = \frac{M_2}{v_1} = \frac{-M_1 + F_1 L}{v_1} = \frac{6EI_z}{(1 + \Phi)L^2}$$

Similarly, if the left-hand end of the beam is clamped, as shown in Fig. 5.23b, then by use of the differential equations for the beam deflections or the condition of symmetry it can be demonstrated that

$$k_{33} = k_{11} = \frac{12EI_z}{(1 + \Phi)L^3}$$

$$k_{43} = -k_{21} = \frac{-6EI_z}{(1 + \Phi)L^2}$$

In order to determine the stiffness coefficients associated with the nodal rotations v'_1 and v'_2 (which are the derivatives of v_b , evaluated at the nodes) the beam is

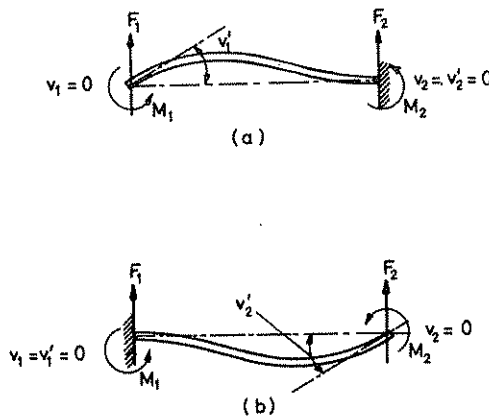


Figure 5.24 Nodal rotations imposed individually.

given a rotation at each end, as shown in Fig. 5.24, and the associated nodal forces and moments are calculated. The force-deflection relationship is again given by (5.5.4), but the constants C_1 and C_2 in these equations must now be evaluated from a different set of boundary conditions. The boundary conditions of Fig. 5.24a are

$$v = 0 \quad \text{at } x = 0, x = L$$

$$\text{and} \quad \frac{dv}{dx} = \frac{dv_s}{dx} = -\frac{F_1}{GA_s} \quad \text{at } x = L$$

Equation 5.5.4 becomes

$$EI_z v = \frac{F_1}{6}(x^3 - L^2 x) + \frac{M_1}{2}(Lx - x^2)$$

and the forces are related by

$$F_1 = \frac{6M_1}{(4 + \Phi)L} \quad (5.5.7)$$

As before, the remaining forces acting on the beam can be determined from the equations of equilibrium, that is, (5.5.6).

Now at $x = 0$

$$\frac{dv_b}{dx} = \frac{dv}{dx} - \frac{dv_s}{dx} = v'_1$$

so that

$$v'_1 = \frac{M_1(1 + \Phi)L}{EI_z(4 + \Phi)} \quad (5.5.8)$$

Hence, from (5.5.6), (5.5.7), and (5.5.8)

$$k_{22} = \frac{M_1}{v'_1} = \frac{(4 + \Phi)EI_z}{(1 + \Phi)L}$$

$$k_{32} = \frac{F_2}{v'_1} = \frac{-F_1}{v'_1} = -\frac{6EI_z}{(1 + \Phi)L^2}$$

$$k_{42} = \frac{M_2}{v'_1} = \frac{-M_1 + F_1 L}{v'_1} = \frac{(2 - \Phi)EI_z}{(1 + \Phi)L}$$

The deflected shape in Fig. 5.24b is the same as in Fig. 5.24a; the figure is simply rotated 180°. Hence it follows that

$$k_{44} = k_{22} = \frac{(4 + \Phi)EI_z}{(1 + \Phi)L}$$

We now summarize these results for the stiffness sub-matrix $\mathbf{k}_o$ corresponding to nodal forces and moments in the xy -plane. For this purpose we define a generalized bending stiffness parameter $\bar{b}_z$ which includes the effect of shear

$$\left. \begin{aligned} \bar{b}_z &= \frac{EI_z}{(1 + \Phi_y)L^3} \\ \text{where } \Phi_y &= \frac{12EI_z}{GA_{sy}L^2} \end{aligned} \right\} \quad (5.5.9)$$

The subscript z indicates that the bending is about the z -axis and the subscript y indicates that the deflection, including the shear deflection, is in the y -direction. A_{sy} is the proportion of the cross-sectional area over which the y -directed shear force is assumed to act. In terms of these quantities the stiffness sub-matrix is

$$\mathbf{k}_o = \bar{b}_z \begin{bmatrix} 12 & & & \text{symm.} \\ 6L & (4 + \Phi_y)L^2 & & \\ -12 & -6L & 12 & \\ 6L & (2 - \Phi_y)L^2 & -6L & (4 + \Phi_y)L^2 \end{bmatrix} \quad (5.5.10)$$

HORIZONTAL DISPLACEMENT AND ROTATION (w AND w')

The stiffness coefficients associated with the displacements in the xz -plane, w and w' , can be derived directly from foregoing results. To distinguish the forces and moments in the two planes the more complete notation F_{y1} , F_{y2} , M_{z1} , and M_{z2} will be used for the nodal forces just derived, that is, forces in the y -direction and moments about the z -direction. Likewise the symbols F_{z1} , F_{z2} , M_{y1} , and M_{y2} will be used for the nodal forces in the xz -plane. It should be observed, however, that with the sign convention of Fig. 5.22 the directions of the positive nodal moments in the yx - and zx -planes are different. This is illustrated in Fig. 5.25 which shows that the positive direction of the nodal moments M_{y1} and M_{y2} is opposite to that of M_{z1} and M_{z2} . From this it can be shown that the following terms in $\mathbf{k}_w$ are of opposite sign to the corresponding terms in $\mathbf{k}_o$: k_{21} , k_{41} , k_{32} , and k_{43} . All other terms have the same sign as in $\mathbf{k}_o$ and therefore

$$\mathbf{k}_w = \bar{b}_y \begin{bmatrix} 12 & & & \text{symm.} \\ -6L & (4 + \Phi_z)L^2 & & \\ -12 & 6L & 12 & \\ -6L & (2 - \Phi_z)L^2 & 6L & (4 + \Phi_z)L^2 \end{bmatrix} \quad (5.5.11)$$

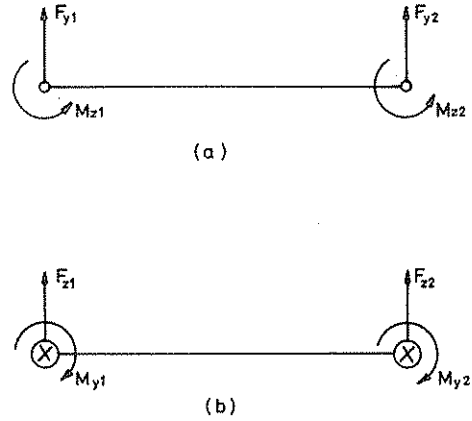


Figure 5.25 Sign convention for nodal forces and moments.

$$\begin{aligned} \text{where } \bar{b}_y &= \frac{EI_y}{(1 + \Phi_z)L^3} \\ \text{and } \Phi_z &= \frac{12EI_y}{GA_{sy}L^2} \end{aligned} \quad (5.5.12)$$

Torsional and Axial Stiffness

The next chapter deals with the torsional loading and response of the hull girder. It includes a summary of the principal aspects of the torsion of prismatic thin-wall beams, and it presents a special beam element that is suitable for structures in which warping of the cross section of the beam is restrained. However, in a free-standing framework made up of slender beams there is little restraint against warping of the beam cross section. In this case the torsional rigidity of the beam is simply GJ , where the torsional stiffness parameter J depends on the shape of the cross section (see Section 6.1 for methods of calculating J). Hence, for the present purpose of deriving a beam element for frame analysis it is appropriate to adopt this simpler expression. Since warping restraint increases the torsional rigidity of a beam any error incurred is on the conservative side. Also, if the frame structure is such that warping restraint could be significant then the hull girder element of Chapter 6 can be used for the analysis; it is not restricted to hull girder applications.

With no warping restraint the differential equation for the twist θ of a beam is

$$M_x = -GJ \frac{d\theta}{dx}$$

Integration gives

$$M_{xx} = -GJ\theta + C_1$$

By imposing a twisting moment M_{x1} at node 1, giving a rotation there of θ_1 , together with the boundary condition $\theta = 0$ at $x = L$, the constant of integration C_1 is found to be

$$C_1 = M_{x1}L$$

Since $\theta = \theta_1$ at $x = 0$, it follows that

$$M_{x1} = \frac{GJ}{L} \theta_1$$

The equilibrium condition for the twisting moment is

$$M_{x2} = -M_{x1}$$

Hence
$$k_{11} = \frac{M_{x1}}{\theta_1} = \frac{GJ}{L}$$

and
$$k_{21} = \frac{M_{x2}}{\theta_1} = \frac{-GJ}{L}$$

From symmetry of the beam element, $k_{22} = k_{11}$. Therefore the stiffness submatrix for the two torsional degrees of freedom θ_1 and θ_2 is

$$\mathbf{k}_\theta = \frac{GJ}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \quad (5.5.13)$$

Finally, from (5.2.1) the axial stiffness submatrix is

$$\mathbf{k}_u = \alpha \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \quad (5.5.14)$$

where

$$\alpha = \frac{AE}{L}$$

Stiffness Matrix for the General Beam Element

Having obtained the four submatrices we can now combine them to form the stiffness matrix for the general beam element as follows

$$\mathbf{k}^e = \begin{bmatrix} \mathbf{k}_u & \text{symm.} \\ 0 & \mathbf{k}_v \\ 0 & 0 & \mathbf{k}_w \\ 0 & 0 & 0 & \mathbf{k}_\theta \end{bmatrix} \quad (5.5.15)$$

in which the four submatrices are given by (5.5.14), (5.5.10), (5.5.11), and (5.5.13).

If it is desired to express $\mathbf{k}^e$ using the degrees of freedom of Fig. 5.22, the various terms of the four submatrices would be inserted into $\mathbf{k}^e$ in the positions indicated by (5.5.1). The resulting stiffness matrix is given in (5.5.16), on page 217.

Coordinate Transformation for a Generally Orientated Beam Element

It is necessary to provide for the case in which the beam has a completely general orientation in three dimensions relative to the structure axes X, Y, Z . For the general beam element the displacement vector at each node contains two sets of components—translation and rotation—and the nodal force vector at each node contains both forces and moments. For this general case the transformation between element axes and structure axes is more conveniently expressed if the degrees of freedom are grouped together at each of the two nodes, as in Fig. 5.22. The transformation matrix is

$$\mathbf{T} = \begin{bmatrix} \lambda & & & \\ & \lambda & & \\ & & \lambda & \\ & & & \lambda \end{bmatrix}$$

where λ is the 3×3 nodal transformation matrix given in (5.2.7).*

*If the principal moments of inertia for the beam elements are different it is also necessary to rotate about the element x -axis.

REFERENCES

1. J. Clarkson, *The Elastic Analysis of Flat Grillages*, Cambridge University Press, 1965.
2. Pin-Yu Chang, A simple method for elastic analysis of grillages, *J. Ship Res.*, **12** (2), June 1968, pp. 153–159.
3. D. Faulkner, Chapter 22 of *Ship Structural Design Concepts*, J. Harvey Evans, Ed., Cornell Maritime Press, 1975.

(5.5.16)

where $\alpha = \frac{EA}{L}$; $\bar{b}_z = \frac{EI_z}{(1 + \Phi_y)L^3}$; $\bar{b}_y = \frac{EI_y}{(1 + \Phi_z)L^3}$; $t = \frac{GJ}{L}$